

Computationally Efficient Estimation of High-Dimensional Exponential Family Models via Tilted Cell Moments and Alternating Projections

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Abstract

Generalized linear models on register-scale data sets combine two structures that jointly defeat standard maximum-likelihood software: a high-dimensional categorical part—detailed demographic strata and one or more fixed-effect factors with thousands to millions of levels—and a low-dimensional continuous part that varies within the resulting cells. When every covariate is categorical, the log likelihood collapses onto cell aggregates and estimation is essentially free; a single continuous covariate destroys this collapse. We develop a framework, *tilted cell moments*, that restores the computational benefit of the cell structure without any approximation: for exponential dispersion families we prove that the exact score vector and information matrix are sums of low-order moments of an exponentially tilted within-cell measure, so that every Newton or Fisher-scoring iteration requires one $O(Np_c^2)$ pass over the microdata, with p_c the number of continuous covariates only, followed by algebra on the J cells. For additional fixed-effect factors we analyze the nonlinear block Gauss–Seidel scheme that alternates closed-form profile updates of the absorbed effects with tilted-moment Newton steps. We prove its global convergence to the maximum-likelihood estimate for canonical links—resolving a gap repeatedly noted in the applied literature—and derive its local linear rate, which for two absorbed factors equals the squared Friedrichs angle cosine between the weighted fixed-effect subspaces and, equivalently, one minus the spectral gap of a weighted bipartite graph Laplacian. We then develop asymptotic theory in three regimes: fixed dimension; a diverging number of cells with $q_N^3/N \rightarrow 0$; and, for the Poisson family, the proportional regime $J/N \rightarrow c \in (0, 1)$ with bounded cell sizes, where a conditional-likelihood representation of the profile likelihood yields consistency, asymptotic normality, and valid cluster-robust inference with no incidental-parameter bias. Extensions cover logistic and gamma models, ℓ_1/ℓ_2 -penalized estimation by proximal

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Newton at cell-collapsed cost, and Firth’s bias-reducing adjustment computed inside the tilted accumulation as a principled response to separation. Simulations confirm that the tilted iterates equal the full-data Newton iterates to machine precision at a small fraction of the cost, that observed convergence rates match the spectral prediction to three decimals, and that nominal 95% intervals attain 94–96% coverage even with one fixed effect per two observations. A register-scale case study with five million person-year episodes and 10,000 absorbed municipality–year effects runs in seconds on a laptop. The framework is implemented in the Stata package `stpois` and in an open-source reference implementation.

Keywords: generalized linear models; exponential tilting; alternating projections; block coordinate ascent; incidental parameters; profile likelihood; separation.

1 Introduction

A large share of applied statistical work now takes place on data sets whose size is set not by a sampling budget but by an administrative process: full-population registers, insurance claim files, electronic health records, digital trace data. Models fitted to such data are rarely exotic—Poisson regression for event rates, logistic regression for binary outcomes, gamma regression for costs—but their covariate structure has a characteristic shape. A small number of continuous covariates of scientific interest (income, exposure, a polygenic score) appear alongside a rich categorical apparatus: demographic strata such as age band $\times$ sex $\times$ education, and one or more fixed-effect factors—region, calendar year, firm, hospital, sequencing batch—whose level counts run from hundreds to millions. The cross-classification of the categorical covariates partitions the N observations into J cells, and it is the interplay between N , J , and the continuous covariates that determines whether maximum-likelihood estimation is a routine computation or an overnight batch job.

Two classical observations frame the problem. First, when *every* covariate is categorical, the exponential-family log likelihood depends on the data only through within-cell aggregates; estimation on the J collapsed cells reproduces the microdata maximum-likelihood estimator (MLE) exactly, and the per-iteration cost drops from $O(Nq^2)$ to $O(Jq^2)$ for q parameters. This is the standard device behind register-based rate modeling in epidemiology and demography (Holford, 1980; Laird and Olivier, 1981; Clayton and Hills, 1993). Second, the collapse is fragile: a single continuous covariate destroys it, because the cumulant function of the family is strictly convex, so cell sums of nonlinear transforms of the linear predictor cannot be recovered from cell means (Section 2.2). Analysts then face an unattractive choice between discretizing the continuous covariates—introducing bias that is hard to quantify—and abandoning the cell structure entirely, paying the full $O(Nq^2)$ price per iteration even though most of the q parameters are attached to covariates that are constant within cells.

A separate literature addresses the fixed-effect factors. For linear models, absorbing high-dimensional fixed effects by within-group demeaning is classical, and the method of alternating projections extends it to multiple factors (Halperin, 1962; Gaure, 2013). For generalized linear models (GLMs), Guimarães and Portugal (2010) introduced an iteratively reweighted scheme whose inner linear problem is solved by weighted alternating projections; refinements are implemented in widely used software: `ppmlhdfe` in Stata (Correia et al., 2020), `fixest` and its predecessor `FENmlm` in R (Bergé, 2018), and `alpaca` (Stammann, 2018). A conspicuous feature of this literature is that the *nonlinear* iteration—the outer loop that alternates fixed-effect updates with covariate updates on the actual GLM likelihood—is used billions of times a year without a convergence guarantee. Correia et al. (2020) prove correctness of the inner linear solves; Bergé (2018) reports that the profile Gauss–Seidel “converges in practice”; the documentation of every major implementation, and the software article accompanying the Stata

command `stpois` that implements the algorithms studied here, state explicitly that formal guarantees for the nonlinear cycle are not available.

This article develops the missing theory and, in doing so, unifies the cell-collapse and fixed-effect-absorption devices into a single framework with quantified computational and statistical properties. Our contributions are fourfold.

First: an exact tilted-moment representation (Section 2). For exponential dispersion families with canonical link we prove (Theorem 1) that the full-model score vector and Fisher information matrix are, without approximation, linear combinations of the zeroth-, first-, and second-order moments of an *exponentially tilted probability measure* on each cell—for the Poisson family, an Esscher transform of the within-cell empirical exposure measure, connecting the construction to the classical tilting literature (Esscher, 1932; Efron, 1981; Jensen, 1995; Kitamura and Stutzer, 1997). Consequently each Newton iteration is computable in a single pass over the microdata whose cost involves only the p_c continuous columns, reducing the per-iteration complexity from $O(N(p_c + k)^2)$ to $O(Np_c^2 + Jk^2)$ (Corollary 1), where k is the number of cell-level parameters. The resulting iterates are *identical* to full-data Newton—not close, identical—which we verify to machine precision in Section 6.1. Fisher scoring under non-canonical links has the same structure (Proposition 2).

Second: convergence theory for nonlinear alternating projections (Section 3). We formulate fixed-effect absorption as exact block coordinate ascent on the (concave, for canonical links) log likelihood, with closed-form profile updates for the absorbed blocks and damped tilted-moment Newton steps for the covariate block. Theorem 2 establishes global convergence to the MLE whenever the MLE exists, under conditions that are sharp in the sense that failure of each corresponds to a recognizable data pathology (separation, disconnected designs). Theorem 3 derives the local convergence rate: the iteration contracts linearly with factor equal to the spectral radius of the block Gauss–Seidel operator of the information matrix at the optimum; for two absorbed factors this radius equals $\cos^2 \theta_F$, the squared cosine of the Friedrichs angle between the weighted fixed-effect subspaces (Aronszajn, 1950; Kayalar and Weinert, 1988; Deutsch, 2001), and Corollary 2 identifies $1 - \cos^2 \theta_F$ with the spectral gap of a weighted bipartite graph built from the two factors—making precise, and quantitative, the folklore that alternating projections slow down when fixed effects are “highly correlated” (Gaure, 2013). Simulations in Section 6.2 match the predicted rates to three decimal places.

Third: asymptotic theory, including high-dimensional regimes (Section 4). Because the tilted-moment estimator *is* the MLE, fixed-dimension consistency, asymptotic normality, and efficiency follow from classical GLM theory (Fahrmeir and Kaufmann, 1985) (Theorem 4). The statistically interesting regimes are high-dimensional. When the total parameter count $q_N = p + k_N$ diverges, we prove consistency in the intrinsic (information) norm and asymptotic normality of fixed contrasts under a balanced-leverage condition and $q_N^3/N \rightarrow 0$ (Theorem 5), by a self-contained argument in the spirit of Portnoy (1988) and He and Shao (2000). In the

proportional regime—one absorbed effect per cell, cells of bounded size, $J/N \rightarrow c \in (0, 1)$ —no such expansion is possible and incidental-parameter bias is the default expectation (Neyman and Scott, 1948; Lancaster, 2000). For the Poisson family we prove that the profile (concentrated) likelihood coincides, up to a constant, with the conditional likelihood obtained by conditioning on cell totals (Lemma 1), a multinomial likelihood with fixed-dimension parameter; Theorem 6 then delivers consistency, asymptotic normality, attainment of the semiparametric efficiency bound for the fixed-effects model, and consistency of both the profile-information and the cell-clustered sandwich variance estimators, all at $J/N \rightarrow c$. The contrast with the logit family, where the profile and conditional likelihoods differ and bias of order $1/m$ persists (Andersen, 1970; Fernández-Val and Weidner, 2016), is discussed. Simulations with one fixed effect per *two* observations show nominal 95% intervals covering at 94–96% (Section 6.3).

Fourth: extensions that inherit the computational structure (Section 5). Logistic and gamma regression fit the framework directly—for the gamma family with log link the Fisher weights are identically one, so the tilted moments reduce to untilted cell moments. Ridge and lasso penalties on the continuous coefficients combine with proximal Newton so that each coordinate-descent sweep costs $O((p_c + k)^2)$, independent of N (Proposition 3). Firth’s bias-reducing adjusted score, the principled remedy for separation, is computable inside the same single pass (Proposition 4), and Section 6.4 shows it resolving quasi-separation that defeats the MLE in 40% of small-sample replications.

Section 7 applies the machinery to a fully reproducible register-scale case study in spatial epidemiology—five million person-year episodes, 10,000 absorbed municipality-year effects, a 96-cell demographic design, and two continuous covariates—which fits, with cluster-robust standard errors, in under half a minute on a laptop, and whose dense-design counterpart would require a terabyte-scale matrix. Section 8 discusses limitations and software. All proofs are in the Appendix; all numerical results are generated by short Python scripts included with the manuscript and re-runnable end to end.

Throughout, the point of view is that of Correia et al. (2020), Gaure (2013), and the broader computational-statistics literature on estimating structured GLMs at scale; what is new here is (i) the exact tilted-moment decomposition and its complexity consequences, (ii) the convergence theory for the nonlinear absorption cycle, and (iii) the high-dimensional asymptotics, in particular the proportional-regime result and its variance theory.

2 Exponential family models and tilted cell moments

2.1 Model and notation

Observations $i = 1, \dots, N$ carry a response y_i , a vector $\mathbf{x}_i \in \mathbb{R}^p$ of *individual-level* covariates (those involving at least one continuous variable, including continuous–categorical interaction columns), a vector $\mathbf{w}_{j(i)} \in \mathbb{R}^k$ of *cell-level* covariates, and a known offset o_i . The map

$j : \{1, \dots, N\} \rightarrow \{1, \dots, J\}$ assigns each observation to the cell of the cross-classification of the categorical covariates; $\mathbf{w}_j$ is constant within cell j by construction. Conditional on the covariates, responses are independent draws from an exponential dispersion family

$$f(y_i | \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}, \quad (1)$$

with $b : \Theta \rightarrow \mathbb{R}$ strictly convex and three times continuously differentiable on the open canonical parameter space $\Theta \subseteq \mathbb{R}$, and dispersion $\phi > 0$ (McCullagh and Nelder, 1989). The mean and variance are $\mu_i = b'(\theta_i)$ and $\text{Var}(y_i) = \phi b''(\theta_i)$. We model the linear predictor

$$\eta_i = \boldsymbol{\gamma}' \mathbf{x}_i + \boldsymbol{\delta}' \mathbf{w}_{j(i)} + o_i, \quad \boldsymbol{\theta} = (\boldsymbol{\gamma}', \boldsymbol{\delta}')' \in \mathbb{R}^q, \quad q = p + k, \quad (2)$$

and, until Section 2.4, take the link to be canonical, $\theta_i = \eta_i$. The Poisson event-history model that motivates much of the applied work is the special case $b(\theta) = e^\theta$, $\phi = 1$, $y_i \in \{0, 1, 2, \dots\}$ an event count, and $o_i = \log t_i$ the log of the time at risk; the piecewise-exponential equivalence of Holford (1980); Laird and Olivier (1981); Whitehead (1980) makes this the canonical likelihood for register-based survival analysis. Up to terms not involving $\boldsymbol{\theta}$, the log likelihood is

$$\ell(\boldsymbol{\theta}) = \frac{1}{\phi} \sum_{i=1}^N \{y_i \eta_i - b(\eta_i)\}. \quad (3)$$

Because (3) is a sum of concave functions of the linear predictor, ℓ is concave on $\mathbb{R}^q$, strictly so on the quotient of $\mathbb{R}^q$ by the null space of the stacked design $\mathbf{a}_i = (\mathbf{x}_i', \mathbf{w}_{j(i)}')'$. We write $\mathbf{U}(\boldsymbol{\theta}) = \nabla \ell$ for the score, $\mathbf{H}(\boldsymbol{\theta}) = -\nabla^2 \ell$ for the negative Hessian, and note that for canonical links $\mathbf{H}$ equals the Fisher information $\mathbf{F}(\boldsymbol{\theta}) = \mathbb{E}[-\nabla^2 \ell]$, so Newton–Raphson and Fisher scoring coincide.

2.2 Cell collapse and its failure

Suppose first that $p = 0$: every covariate is categorical, so $\eta_i = \boldsymbol{\delta}' \mathbf{w}_j + o_i$ varies over i within a cell only through the offset. For the Poisson family with $o_i = \log t_i$, (3) becomes $\ell(\boldsymbol{\delta}) = \sum_j \{D_j \boldsymbol{\delta}' \mathbf{w}_j - T_j e^{\boldsymbol{\delta}' \mathbf{w}_j}\} + \text{const}$, with $D_j = \sum_{i \in j} y_i$ and $T_j = \sum_{i \in j} t_i$: the collapsed cell data (D_j, T_j) carry all the information, and estimation on the J cells is exact. The following proposition records why no analogous finite collapse survives the introduction of a continuous covariate, and in what precise sense. Define the within-cell *exposure measure* $\nu_j = \sum_{i \in j} e^{o_i} \delta_{\mathbf{x}_i}$ on $\mathbb{R}^p$ (for Poisson, $\nu_j = \sum_{i \in j} t_i \delta_{\mathbf{x}_i}$), a finite discrete measure recording the joint configuration of exposures and continuous covariates in cell j .

Proposition 1 (Collapse and its failure). *Consider the Poisson family with offsets. (i) If $\mathbf{x}_i = \bar{\mathbf{x}}_j$ for all $i \in j$ and all j , the log likelihood (3) depends on the data only through $\{(D_j, T_j, \bar{\mathbf{x}}_j, \sum_{i \in j} y_i \bar{\mathbf{x}}_i)\}_{j \leq J}$, and the MLE on the collapsed cells equals the microdata MLE. (ii)*

In general, $\ell(\boldsymbol{\gamma}, \boldsymbol{\delta})$ depends on the data through $\{S_j, D_j, \mathcal{L}_{\nu_j}\}_{j \leq J}$, where $S_j = \sum_{i \in j} y_i \mathbf{x}_i$ and $\mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) = \int e^{\boldsymbol{\gamma}' \mathbf{x}} \nu_j(d\mathbf{x}) = \sum_{i \in j} t_i e^{\boldsymbol{\gamma}' \mathbf{x}_i}$ is the Laplace transform of ν_j ; and $\mathcal{L}_{\nu_j}$ determines ν_j . Consequently any statistic from which the likelihood function can be computed for all $\boldsymbol{\gamma}$ in an open set must determine the entire within-cell exposure measure, whose support size—hence the dimension of any such statistic—grows with the number of distinct covariate values in the cell. (iii) By strict convexity of the exponential (Jensen), $\mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) \geq T_j e^{\boldsymbol{\gamma}' \bar{\mathbf{x}}_j}$ with equality for some $\boldsymbol{\gamma} \neq \mathbf{0}$ direction only if $\mathbf{x}$ is ν_j -a.s. constant along that direction: replacing microdata by cell means strictly understates the within-cell mass of the cumulant term and is not a reparametrization of the same likelihood.

Part (ii) is the formal statement of the practical dictum that “continuous covariates destroy collapsibility”: the fixed-dimensional sufficient statistic $\sum_i y_i \mathbf{a}_i$ for the canonical parameter still exists, but *computing the likelihood*—and hence the MLE—requires functionals of the data whose dimension is of order N , not J . The next subsection shows that the required functionals nevertheless have a special structure: for each trial value of $\boldsymbol{\gamma}$ they are three low-order moments of a tilted measure, computable in one linear pass.

2.3 Exponentially tilted cell moments

Definition 1 (Tilted within-cell measures). *Fix $\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\delta})$ and let $\omega_i = b''(\eta_i) > 0$ be the information weight of observation i . The information-tilted measure on cell j is the probability measure*

$$P_{j,\boldsymbol{\theta}}(\{i\}) = \frac{\omega_i}{\sum_{i' \in j} \omega_{i'}}, \quad i \in j.$$

For the Poisson family, $\omega_i = \mu_i = t_i e^{\boldsymbol{\delta}' \mathbf{w}_j + \boldsymbol{\gamma}' \mathbf{x}_i}$, the factor $e^{\boldsymbol{\delta}' \mathbf{w}_j}$ cancels, and $P_{j,\boldsymbol{\theta}}$ reduces to the Esscher transform (exponential tilt) of the normalized exposure measure ν_j/T_j :

$$P_{j,\boldsymbol{\gamma}}(d\mathbf{x}) = \frac{e^{\boldsymbol{\gamma}' \mathbf{x}} \nu_j(d\mathbf{x})}{\mathcal{L}_{\nu_j}(\boldsymbol{\gamma})}, \quad (4)$$

free of $\boldsymbol{\delta}$. Its moments are derivatives of the within-cell cumulant generating function: with $K_j(\boldsymbol{\gamma}) = \log \mathcal{L}_{\nu_j}(\boldsymbol{\gamma})$, $\mathbb{E}_{P_{j,\boldsymbol{\gamma}}}[\mathbf{x}] = \nabla K_j(\boldsymbol{\gamma})$ and $\text{Var}_{P_{j,\boldsymbol{\gamma}}}(\mathbf{x}) = \nabla^2 K_j(\boldsymbol{\gamma})$.

The tilt (4) is precisely the change of measure that underlies saddlepoint approximation (Daniels, 1954; Jensen, 1995) and exponential-tilting estimators in the empirical-likelihood family (Efron, 1981; Kitamura and Stutzer, 1997; Schennach, 2007); here it appears not as an approximation device but as an exact algebraic identity for the score and information, as follows. Write, for each cell j ,

$$M_j^{(0)} = \sum_{i \in j} \omega_i, \quad \mathbf{m}_j = \sum_{i \in j} \omega_i \mathbf{x}_i, \quad \mathbf{Q} = \sum_{i=1}^N \omega_i \mathbf{x}_i \mathbf{x}_i', \quad (5)$$

and let $B_j = \sum_{i \in j} b'(\eta_i)$ and $R_j = \sum_{i \in j} (y_i - b'(\eta_i))$ denote the cell sums of means and of raw residuals.

Theorem 1 (Exact tilted-moment representation). *For the model (1)–(2) with canonical link, at every $\boldsymbol{\theta} \in \mathbb{R}^q$:*

(i) *The score is*

$$\phi \mathbf{U}(\boldsymbol{\theta}) = \begin{pmatrix} \sum_i y_i \mathbf{x}_i - \sum_j \mathbf{m}_j^{b'} \\ \sum_j \mathbf{w}_j R_j \end{pmatrix}, \quad \mathbf{m}_j^{b'} = \sum_{i \in j} b'(\eta_i) \mathbf{x}_i = B_j \mathbb{E}_{P_{j,\boldsymbol{\theta}}^{b'}}[\mathbf{x}],$$

where $P_{j,\boldsymbol{\theta}}^{b'}$ is the tilt of Definition 1 with b' in place of b'' .

(ii) *The information matrix is*

$$\phi \mathbf{F}(\boldsymbol{\theta}) = \begin{pmatrix} \mathbf{Q} & \sum_j \mathbf{m}_j \mathbf{w}_j' \\ \sum_j \mathbf{w}_j \mathbf{m}_j' & \sum_j M_j^{(0)} \mathbf{w}_j \mathbf{w}_j' \end{pmatrix}, \quad \mathbf{m}_j = M_j^{(0)} \mathbb{E}_{P_{j,\boldsymbol{\theta}}}[\mathbf{x}], \quad \mathbf{Q} = \sum_j M_j^{(0)} \mathbb{E}_{P_{j,\boldsymbol{\theta}}}[\mathbf{x} \mathbf{x}'].$$

(iii) *For the Poisson family, $b' = b''$, the two tilts coincide with the $\boldsymbol{\delta}$ -free Esscher transform (4), and $B_j = M_j^{(0)} = e^{\boldsymbol{\delta}' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\boldsymbol{\gamma})$, $\mathbf{m}_j = M_j^{(0)} \nabla K_j(\boldsymbol{\gamma})$, and the $\boldsymbol{\gamma}$ -block of $\mathbf{F}$ is $\sum_j M_j^{(0)} \{\nabla^2 K_j(\boldsymbol{\gamma}) + \nabla K_j \nabla K_j'\}$.*

(iv) *The Newton step $\boldsymbol{\theta}^+ = \boldsymbol{\theta} + \mathbf{F}(\boldsymbol{\theta})^{-1} \mathbf{U}(\boldsymbol{\theta})$ computed from the quantities in (i)–(ii) is identical to the Newton step computed from the full $N \times q$ design.*

Claim (iv) is trivial once (i)–(ii) are established—the two computations produce the same matrices—but it is the operative point: the decomposition involves *no linearization, discretization, or approximation of any kind*, so the accelerated algorithm is not a fast approximate estimator but a fast exact organization of the arithmetic of the MLE. Section 6.1 exhibits agreement of the full iterate *sequences* at the 10^{-13} level, which is the accumulation of floating-point reordering only.

Statistically, the representation has content beyond bookkeeping. The $\boldsymbol{\delta}$ -score in (i) compares each cell's observed event mass with its expected mass under the current tilt; the cross-information in (ii) is the tilted covariance between cell-level and individual-level design; and the $\boldsymbol{\gamma}$ -block in (iii) decomposes into a within-cell tilted covariance $\nabla^2 K_j$ plus a between-cell term. In particular the identification of $\boldsymbol{\gamma}$ rests exactly on $\sum_j M_j^{(0)} \nabla^2 K_j(\boldsymbol{\gamma}) \succ \mathbf{0}$: continuous covariates are identified through their *within-cell, tilt-weighted* variation, a fact that the block structure displays and that drives the Schur-complement profile Hessian used throughout Sections 3–4.

Corollary 1 (Per-iteration complexity). *One Newton iteration computed via Theorem 1 costs*

$$\underbrace{O(N(p^2 + p + 1))}_{\text{microdata pass}} + \underbrace{O(J(k^2 + kp))}_{\text{cell assembly}} + \underbrace{O((p + k)^3)}_{\text{solve}}$$

floating-point operations and one $O(N)$ memory sweep, versus $O(N(p+k)^2) + O((p+k)^3)$ for the full-design iteration. When $N \gg J(p+k)$ the leading-order saving is the factor $(p+k)^2/(p^2+p+1)$, which grows quadratically in the richness k of the categorical structure; when additionally $p = 0$ the microdata pass degenerates to the classical cell collapse and the entire iteration runs at $O(Jk^2)$.

Two remarks on the constant factors. First, the microdata pass is a fused stream: one evaluation of $b'(\eta_i)$, $b''(\eta_i)$ per observation, $p+1$ multiply–accumulate reductions into cell bins, and a rank-one update of the $p \times p$ Gram matrix; on modern hardware it is memory-bandwidth-bound, so measured speedups (Section 6.1) are large but smaller than the FLOP ratio. Second, the accumulations are plain segmented sums—`bincount` operations—so they parallelize and stream trivially; no $N \times q$ design matrix is ever formed, which converts a memory-infeasible problem (Section 7) into an in-cache one.

2.4 Non-canonical links

For a non-canonical, three-times-differentiable, strictly monotone link g with $\mu_i = g^{-1}(\eta_i)$, define the usual Fisher weight and working residual

$$\omega_i = \frac{1}{V(\mu_i) g'(\mu_i)^2}, \quad r_i = (y_i - \mu_i) g'(\mu_i) \omega_i,$$

with V the variance function. Fisher scoring replaces $(b'(\eta_i), b''(\eta_i))$ by (μ_i, ω_i) and the raw residual by r_i .

Proposition 2 (Tilted Fisher scoring). *With ω_i and r_i as above, the expected-information scoring step admits the representation of Theorem 1 with $M_j^{(0)}, \mathbf{m}_j, \mathbf{Q}$ built from ω_i and with $R_j = \sum_{i \in j} r_i$; the fixed points of the resulting iteration are exactly the roots of the ML score equations, and Corollary 1 applies verbatim. For the gamma family with log link, $V(\mu) = \mu^2$ and $g'(\mu) = 1/\mu$ give $\omega_i \equiv 1$: the tilted moments reduce to untilted cell moments of the covariates, fixed across iterations, so after the first pass each subsequent iteration costs $O(Np) + O(Jk^2)$ only.*

3 High-dimensional fixed effects and nonlinear alternating projections

3.1 Profile formulation

Beyond the cell design, applications routinely require one or more fixed-effect factors whose level counts preclude even cell-level treatment: firms, hospitals, zip-code–year interactions.

Algorithm 1 Tilted-moment Newton with absorbed factors (nonlinear block Gauss–Seidel)

```
1: Initialize  $\beta_0, \alpha_0 = \mathbf{0}$ ; drop observations in absorbed groups with no events (Section 3.4).
2: repeat
3:   repeat ▷ inner loop: exact block maximization over  $\alpha$ 
4:     for  $g = 1, \dots, G$  do
5:       update  $\alpha^{(g)}$  by the closed-form profile rule (7)
6:     end for
7:   until  $\max_g \|\Delta \alpha^{(g)}\|_\infty < \tau_{\text{in}}$ 
8:   accumulate tilted moments (5) at  $(\beta_t, \alpha_t)$  in one microdata pass
9:    $\beta_{t+1} \leftarrow \beta_t + \lambda_t \mathbf{F}^{-1} \mathbf{U}$  with  $\lambda_t \in \{1, \frac{1}{2}, \frac{1}{4}, \dots\}$  chosen by backtracking (Armijo) on
      $\ell(\cdot, \hat{\alpha}(\cdot))$ 
10: until  $\max_\ell |\Delta \beta_\ell| / (1 + |\beta_\ell|) < \tau$ 
```

Augment the linear predictor with G absorbed factors,

$$\eta_i = \gamma' \mathbf{x}_i + \boldsymbol{\delta}' \mathbf{w}_{j(i)} + \sum_{g=1}^G \alpha_{g(i)}^{(g)} + o_i, \quad (6)$$

where factor g has J_g levels with level map $g(i)$ and effect vector $\alpha^{(g)} \in \mathbb{R}^{J_g}$, and $J_1 + \dots + J_G$ may be of order N . The full parameter is $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\alpha})$ with $\boldsymbol{\beta} = (\boldsymbol{\gamma}, \boldsymbol{\delta})$. The design has a known null space $\mathcal{K} \subset \mathbb{R}^{q + \sum_g J_g}$ (at least the G dimensions of shifts between the factors' levels and the intercept, more if the factor graph is disconnected); all statements below are on the quotient $\mathbb{R}^{q + \sum J_g} / \mathcal{K}$, realized in practice by a normalization. The estimator of interest is the MLE of $\boldsymbol{\beta}$ with $\boldsymbol{\alpha}$ concentrated out: the profile likelihood $\ell_p(\boldsymbol{\beta}) = \max_{\boldsymbol{\alpha}} \ell(\boldsymbol{\beta}, \boldsymbol{\alpha})$.

For the Poisson family the inner maximization has a closed-form coordinate solution: given all other parameters, the optimal effect of level a of factor g is

$$\alpha_a^{(g)} = \log \frac{\sum_{i:g(i)=a} y_i}{\sum_{i:g(i)=a} \exp(\eta_i - \alpha_a^{(g)})}, \quad (7)$$

the log of the ratio of observed to expected-at-zero-effect events, which is the multiplicative-update rule used by `FENmlm/fixest` (Bergé, 2018) and, in IRLS-linearized form, by `ppmlhdfc` (Correia et al., 2020). Cycling (7) over factors is a nonlinear block Gauss–Seidel scheme; combined with tilted-moment Newton steps for $\boldsymbol{\beta}$ it yields Algorithm 1.

Algorithm 1 is, up to the tilted-moment acceleration of the $\boldsymbol{\beta}$ -step, the algorithm used across the applied HDFE–GLM software ecosystem, and it is the algorithm for which the literature records no convergence guarantee: Correia et al. (2020) prove convergence of the *linear* inner solves they employ, Bergé (2018) and the `stpois` software article report empirical convergence of the nonlinear cycle, and Gaure (2013) analyzes the linear case. The next two subsections close this gap.

3.2 Global convergence

We impose:

Assumption 1. *The link is canonical and b is strictly convex and twice continuously differentiable on $\Theta = \mathbb{R}$.*

Assumption 2. *The MLE exists: $\sup_{\theta} \ell$ is attained at some θ^* (unique modulo $\mathcal{K}$).*

Assumption 3. *(i) Each absorbed group contains at least one observation with y_i in the interior of the convex hull of the support of the family (for Poisson and binomial: at least one $y_i > 0$, and for binomial also one $y_i < n_i$), so that each one-dimensional block likelihood in (7) attains a finite maximum; (ii) the β -block information $\mathbf{F}_{\beta\beta}(\theta)$ is nonsingular on the initial superlevel set $\{\ell \geq \ell(\theta_0)\}$.*

Assumption 2 is a data condition—absence of separation—with a long literature of verifiable characterizations (Haberman, 1974; Wedderburn, 1976; Albert and Anderson, 1984; Santos Silva and Tenreiro, 2010; Correia et al., 2019); Section 3.4 discusses enforcement and the Firth alternative when it fails. Assumption 3(i) is enforced constructively by the drop rule of Algorithm 1, mirroring standard practice (Correia et al., 2020).

Theorem 2 (Global convergence of nonlinear alternating projections). *Under Assumptions 1–3, the iterates $\theta_t = (\beta_t, \alpha_t)$ of Algorithm 1 satisfy: (i) $\ell(\theta_t)$ is nondecreasing and converges to $\sup \ell = \ell(\theta^*)$; (ii) $[\theta_t] \rightarrow [\theta^*]$ in the quotient $\mathbb{R}^{q+\sum J_g}/\mathcal{K}$; in particular $\beta_t \rightarrow \beta^*$ whenever the normalization places the flat directions in the absorbed blocks; and (iii) the same conclusions hold if the β -update is replaced by exact block maximization, and for any cyclic or essentially-cyclic ordering of the blocks.*

The proof (Appendix B) proceeds by establishing that strict concavity on the quotient plus attainment forces compact superlevel sets (Lemma 2); monotonicity then confines the iterates to a compact set, every limit point is shown to be a blockwise maximizer using the uniqueness of the block updates and the sufficient-ascent property of the damped Newton step, and for smooth concave functions blockwise maximizers are global maximizers. The argument is in the tradition of block-coordinate methods (Bertsekas, 1999; Tseng, 2001); what requires care is (a) the quotient structure (the likelihood is not coercive in the natural parametrization, only modulo $\mathcal{K}$), (b) the inexact β -block (a damped Newton step, not an exact maximizer), and (c) dispensing with any compactness or boundedness assumption on the parameter space, which would be false here and is where informal appeals to standard results break down.

Theorem 2 has a practical corollary worth stating: since every accumulation point of the algorithm is *the* MLE, the familiar diagnostics of the linear-projection literature (sensitivity of estimates to tightened tolerances, discrepancies across implementations) reflect finite-iteration error only, not multiplicity; and step-halving on the profile objective—re-solving the inner loop at every trial point, as Algorithm 1 specifies—is guaranteed to terminate at each outer iteration.

3.3 Local rate: Friedrichs angles and a bipartite spectral gap

Convergence speed is governed by the geometry of the absorbed subspaces. Let $\Omega^* = \text{diag}(\omega_i^*)$ collect the information weights at the optimum, let $V_g = \text{span}\{\text{indicator columns of factor } g\} \subset \mathbb{R}^N$, and equip $\mathbb{R}^N$ with the inner product $\langle u, v \rangle_{\Omega^*} = u' \Omega^* v$. Recall that the *Friedrichs angle* $\theta_F \in (0, \pi/2]$ between closed subspaces V_1, V_2 with intersection V_0 has cosine

$$c_F = \sup \left\{ \frac{\langle u, v \rangle_{\Omega^*}}{\|u\|_{\Omega^*} \|v\|_{\Omega^*}} : u \in V_1 \cap V_0^\perp, v \in V_2 \cap V_0^\perp, u, v \neq \mathbf{0} \right\} < 1.$$

Theorem 3 (Local linear convergence). *Let Assumptions 1–3 hold and let b be three times continuously differentiable. (i) The inner iteration of Algorithm 1 at fixed β (the map $T : \alpha \mapsto \alpha^+$ of one Gauss–Seidel sweep of exact block maximizations) is continuously differentiable in a neighborhood of its fixed point $\hat{\alpha}(\beta)$, with Jacobian equal to the block Gauss–Seidel iteration matrix of the α -block information $\mathbf{F}_{\alpha\alpha}$ at the fixed point; consequently the inner iteration converges locally R -linearly with rate $\rho_{\text{GS}} = \rho((\mathbf{D} + \mathbf{L})^{-1} \mathbf{L}') < 1$ on the quotient, where $\mathbf{F}_{\alpha\alpha} = \mathbf{D} + \mathbf{L} + \mathbf{L}'$ is the block decomposition into diagonal and strictly lower/upper block-triangular parts. (ii) For $G = 2$ absorbed factors,*

$$\rho_{\text{GS}} = c_F^2,$$

the squared Friedrichs cosine between V_1 and V_2 in the Ω^ inner product; the same rate governs the weighted alternating-projections demeaning of any working vector, by the theorem of Aronszajn (1950) and Kayalar and Weinert (1988). (iii) The outer iteration (full cycles of Algorithm 1 with exact β -maximization) converges locally R -linearly with rate at most $\rho(\text{block GS operator of } \mathbf{F}(\theta^*) \text{ in the ordering } (\alpha^{(1)}, \dots, \alpha^{(G)}, \beta)) < 1$.*

Corollary 2 (Bipartite spectral gap). *For $G = 2$, define the weighted biadjacency matrix $\mathbf{B} \in \mathbb{R}^{J_1 \times J_2}$ by $B_{ab} = \sum_{i: g_1(i)=a, g_2(i)=b} \omega_i^*$, and let $\mathbf{D}_1 = \text{diag}(\mathbf{B}\mathbf{1})$, $\mathbf{D}_2 = \text{diag}(\mathbf{B}'\mathbf{1})$. Then*

$$c_F = \sigma_2(\mathbf{D}_1^{-1/2} \mathbf{B} \mathbf{D}_2^{-1/2}),$$

the second-largest singular value of the normalized biadjacency (transition) kernel. Equivalently, $1 - c_F^2$ is the spectral gap of the random walk on the weighted bipartite graph of the two factors, and $c_F < 1$ if and only if that graph is connected—the classical identification condition for two-way fixed effects (Abowd et al., 1999; Gaure, 2013).

Corollary 2 turns the folklore statement “correlated fixed effects make demeaning slow” into a computable quantity: before iterating at all, σ_2 of a $J_1 \times J_2$ sparse matrix (a few Lanczos iterations) predicts the per-sweep contraction factor exactly. It also gives sharp intuition for the pathologies: nearly disconnected mobility graphs—workers who rarely switch firms, patients who rarely switch hospitals—have $\sigma_2 \uparrow 1$ and predictably slow demeaning, while balanced

crossed designs have small σ_2 and converge in a handful of sweeps. Section 6.2 verifies the prediction to three decimals across two orders of magnitude of $1 - c_F^2$.

3.4 Separation

Assumption 2 fails in the maximum-likelihood sense when a hyperplane in covariate space separates events from non-events: components of $\hat{\boldsymbol{\theta}}$ diverge and the supremum is not attained (Albert and Anderson, 1984; Santos Silva and Tenreiro, 2010). In the absorbed blocks the failure mode is transparent—a group with no events has $\hat{\alpha}_a = -\infty$ by (7)—and Algorithm 1 drops such groups a priori, after which Assumption 3(i) holds by construction; this mirrors Correia et al. (2019), who treat detection in full generality. Separation induced by the β -block cannot be dropped away and surfaces as non-convergence. Section 5.3 develops the principled remedy—Firth’s adjusted score inside the tilted accumulation—and Section 6.4 quantifies its behavior.

4 Asymptotic theory

Because Theorem 1 makes the tilted-moment estimator *equal* to the MLE, its asymptotic analysis is the analysis of the MLE in the relevant regime. We give results in three regimes, in increasing order of dimensionality. Throughout, $\boldsymbol{\theta}_0$ denotes the truth, expectations are under the model, and the triangular-array framework lets all design quantities depend on N .

4.1 Fixed dimension

Assumption 4. (i) $q = p + k$ is fixed; covariates and offsets are uniformly bounded; (ii) $\lambda_{\min}(\mathbf{F}_N(\boldsymbol{\theta}_0)) \rightarrow \infty$ and $N^{-1}\mathbf{F}_N(\boldsymbol{\theta}_0) \rightarrow \mathbf{F}_\infty \succ \mathbf{0}$; (iii) $\boldsymbol{\theta}_0$ is interior to the parameter space and b is three times continuously differentiable.

Theorem 4 (Classical asymptotics). *Under Assumptions 1 and 4, with probability tending to one the MLE $\hat{\boldsymbol{\theta}}_N$ exists, is unique, and satisfies $\sqrt{N}(\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \phi \mathbf{F}_\infty^{-1})$; the estimator is asymptotically efficient, attaining the Cramér–Rao bound. If the conditional mean is correctly specified but the distribution is not, the same limit holds with sandwich variance $\mathbf{F}_\infty^{-1} \boldsymbol{\Sigma} \mathbf{F}_\infty^{-1}$, consistently estimated by the cell-accumulated sandwich of Section 5.4.*

The proof (Appendix C) is a specialization of Fahrmeir and Kaufmann (1985) plus the pseudo-ML results of Gouriéroux et al. (1984); we include it because the triangular-array statement with offsets and the exact form of the cell-accumulated sandwich are used repeatedly later.

4.2 Diverging dimension: $q_N^3/N \rightarrow 0$

Register applications place $k_N = \dim(\boldsymbol{\delta})$ in the hundreds or thousands—period $\times$ region $\times$ education cells—so the fixed- q regime is a poor description. Let $q_N = p + k_N \rightarrow \infty$, write $\mathbf{F}_N = \mathbf{F}_N(\boldsymbol{\theta}_0)$, and define the *leverage* of observation i as $h_i = \omega_i \mathbf{a}_i' \mathbf{F}_N^{-1} \mathbf{a}_i$ (so $\sum_i h_i = q_N$).

Assumption 5. (i) *Covariates and offsets are uniformly bounded, and the true linear predictors $\{\eta_{0i}\}$ lie in a fixed compact set;* (ii) *balanced leverage: $\max_i h_i \leq C q_N/N$ for a fixed constant C ;* (iii) $q_N^3/N \rightarrow 0$; (iv) b *is three times continuously differentiable with b'' bounded away from 0 and ∞ , and b''' bounded, on compact sets.*

Condition (ii) is the natural design condition: for a pure cell design with balanced cells it holds with C close to 1, and it fails exactly when some cell is drastically undersized relative to the average—the same configurations for which practitioners distrust the asymptotics.

Theorem 5 (Diverging dimension). *Under Assumptions 1 and 5, with probability tending to one the MLE exists and satisfies*

$$\|\mathbf{F}_N^{1/2}(\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}_0)\| = O_p(\sqrt{q_N}),$$

and for any fixed-dimensional matrix of contrasts $\mathbf{C}_N' \in \mathbb{R}^{d \times q_N}$ with $\mathbf{C}_N' \mathbf{F}_N^{-1} \mathbf{C}_N \rightarrow \boldsymbol{\Sigma}_C \succ \mathbf{0}$,

$$(\mathbf{C}_N' \mathbf{F}_N^{-1} \mathbf{C}_N)^{-1/2} \mathbf{C}_N'(\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \phi \mathbf{I}_d).$$

The plug-in variance estimator $\mathbf{C}_N' \mathbf{F}_N(\hat{\boldsymbol{\theta}}_N)^{-1} \mathbf{C}_N$ is consistent in relative terms.

The proof (Appendix C.2) is self-contained and short by the standards of this literature, because concavity in the canonical parametrization substitutes for the empirical-process machinery needed in general M -estimation (Portnoy, 1988; He and Shao, 2000; Wang, 2011): a quadratic-minorization argument in the intrinsic norm delivers the rate, and a third-order Taylor control with the balanced-leverage condition delivers asymptotic linearity with remainder $o_p(1)$ exactly when $q_N^3/N \rightarrow 0$.

4.3 The proportional regime $J/N \rightarrow c$: Poisson profile likelihood as conditional likelihood

The regime that motivates this paper’s title question is more extreme: one absorbed effect per cell with cells of *bounded* size, so the parameter count is a positive fraction of the sample size and Theorem 5 cannot apply ($q_N \asymp N$). This is the incidental-parameters regime of Neyman and Scott (1948); for smooth nonlinear panel models the profile MLE of the common parameter is typically inconsistent or biased at order $1/m$ for cell size m (Lancaster, 2000; Fernández-Val and Weidner, 2016). The Poisson family with log link is a distinguished exception, for a structural reason worth displaying: profiling and conditioning coincide.

Consider cells $j = 1, \dots, J$ of sizes $m_j \leq \bar{m} < \infty$, one free effect α_j per cell, individual-level covariates $\mathbf{x}_{ij}$, offsets o_{ij} , and $y_{ij} \sim \text{Poisson}(\exp(\alpha_j + \gamma' \mathbf{x}_{ij} + o_{ij}))$ independently. Let $D_j = \sum_i y_{ij}$.

Lemma 1 (Profile equals conditional). *For every γ with all D_j -positive cells,*

$$\ell_p(\gamma) = \max_{\alpha} \ell(\gamma, \alpha) = \sum_{j: D_j > 0} \log \mathbb{P}(y_{j1}, \dots, y_{jm_j} \mid D_j; \gamma) + \text{terms not involving } \gamma,$$

where the conditional law of $(y_{j1}, \dots, y_{jm_j})$ given D_j is multinomial with D_j trials and probabilities $\pi_{ij}(\gamma) = e^{\gamma' \mathbf{x}_{ij} + o_{ij}} / \sum_{i'} e^{\gamma' \mathbf{x}_{i'j} + o_{i'j}}$. Moreover ℓ_p is concave in γ , and its score and negative Hessian are the Schur complements of the tilted moments:

$$\mathbf{U}_p(\gamma) = \sum_j \sum_i (y_{ij} - D_j \pi_{ij}) \mathbf{x}_{ij}, \quad \mathbf{H}_p(\gamma) = \sum_j D_j \text{Var}_{\pi_j}(\mathbf{x}_j) = \mathbf{Q} - \sum_j \frac{\mathbf{m}_j \mathbf{m}_j'}{M_j^{(0)}}.$$

Lemma 1 is a fixed-effects analogue of the classical Poisson–multinomial connection; versions of the first display appear in the econometric panel literature (Andersen, 1970; Hausman et al., 1984; Blundell et al., 2002; Wooldridge, 1999), and the Schur-complement identity is what the tilted-moment machinery computes at $O(Np^2)$ per iteration (function `profile_poisson` in the accompanying code). Its consequence, however, seems not to have been developed as an explicit high-dimensional limit theorem with variance theory, which we now state. Conditional on covariates and cell effects, cells are independent; define the average conditional information $\bar{\mathbf{F}}_J(\gamma_0) = J^{-1} \sum_j \mathbb{E}[D_j \text{Var}_{\pi_j}(\mathbf{x}_j)]$.

Assumption 6. (i) Cell sizes satisfy $2 \leq m_j \leq \bar{m}$; $J/N \rightarrow c \in (0, 1)$; (ii) covariates and offsets are uniformly bounded and $\sup_j \alpha_j \leq C$, so cell means are bounded above (they may be arbitrarily small); (iii) $\bar{\mathbf{F}}_J(\gamma_0) \rightarrow \bar{\mathbf{F}} \succ \mathbf{0}$ as $J \rightarrow \infty$; (iv) γ_0 is interior to a fixed compact set Γ .

Theorem 6 (Proportional regime). *Under Assumption 6: (i) the profile MLE $\hat{\gamma}_J$ is consistent; (ii)*

$$\sqrt{J} (\hat{\gamma}_J - \gamma_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \bar{\mathbf{F}}^{-1}),$$

with no asymptotic bias, despite the number of nuisance parameters being a positive fraction of N ; (iii) both the profile information estimator $J \mathbf{H}_p(\hat{\gamma}_J)^{-1}$ and the cell-clustered sandwich estimator $J \mathbf{H}_p^{-1}(\frac{G}{G-1} \sum_j \mathbf{s}_j \mathbf{s}_j') \mathbf{H}_p^{-1}$, with $\mathbf{s}_j$ the per-cell profile score, are consistent for the asymptotic variance in (ii)—the latter remaining consistent under within-cell dependence misspecification with correctly specified conditional means; and (iv) $\hat{\gamma}_J$ attains the semiparametric efficiency bound for the model in which the α_j are unrestricted nuisance parameters, the conditional score being the efficient score.

Three comments. First, the mechanism is special to the Poisson–log pair: the profile score is *exactly* the conditional score, hence exactly unbiased for every J and m ; the incidental-parameter

bias that plagues, e.g., the fixed-effects logit (where conditional ML exists but differs from profile ML, and profile ML has $O(1/m)$ bias (Andersen, 1970; Fernández-Val and Weidner, 2016)) is identically zero here. This is the likelihood-theoretic content of the folklore that “Poisson FE is immune to the incidental-parameters problem” (Lancaster, 2000; Wooldridge, 1999), sharpened to a $J/N \rightarrow c$ limit theorem with variance estimation. Second, the theorem’s variance is the *conditional* information, which the naive full-information plug-in would overstate; the tilted-moment machinery produces the correct Schur complement at no extra cost. Third, event-free cells contribute no information and are dropped; this is the profile-likelihood counterpart of the separation drop rule and is accounted for in the theory by summing over $D_j > 0$.

Section 6.3 probes the theorem at $c = 1/2$ —one fixed effect per two observations—where nominal 95% intervals from both estimators in (iii) cover at 94–96%.

5 Extensions

5.1 Families beyond Poisson

The framework of Section 2 was stated for a generic exponential dispersion family; we record the two most used instances.

Logistic regression. With $b(\theta) = \log(1 + e^\theta)$, $\mu_i = (1 + e^{-\eta_i})^{-1}$ and $\omega_i = \mu_i(1 - \mu_i)$: the tilt weights each within-cell observation by its Bernoulli variance, so the tilted measure concentrates on observations near the decision boundary. All of Theorem 1, Corollary 1, and Theorems 2–3 apply verbatim (the logit link is canonical). Unlike the Poisson case the tilt depends on δ through ω_i , so the within-cell moments must be re-accumulated each iteration—which the single pass does anyway. The proportional-regime theory of Section 4.3 does *not* transfer: profiling and conditioning differ for the logit, conditional ML requires the permanent-type sums of Andersen (1970), and profile ML carries $O(1/m)$ incidental-parameter bias (Fernández-Val and Weidner, 2016); the honest options at $J/N \rightarrow c$ are conditional ML (computable but heavier) or bias correction, and we view the integration of either into the tilted-moment pass as valuable future work.

Gamma regression with log link. Here the log link is not canonical, Proposition 2 applies, and the Fisher weights are identically one: the “tilted” moments are ordinary exposure-weighted cell moments, constant across iterations. One pass suffices for the whole optimization, and only the working residual sums R_j are refreshed—a family where the acceleration is at its most extreme.

5.2 Penalized estimation

In applications with many continuous covariates (genomics: expression levels alongside batch cells), sparsity-inducing penalties on γ are natural while leaving δ and the absorbed effects

unpenalized. Consider

$$\min_{\gamma, \delta} -\ell(\gamma, \delta) + \lambda_1 \|\gamma\|_1 + \frac{\lambda_2}{2} \|\gamma\|_2^2.$$

Proposition 3 (Penalized tilted-moment proximal Newton). *At each outer iteration, form the quadratic model of $-\ell$ from the tilted moments (5) (one $O(Np^2)$ pass), i.e., the exact second-order expansion with gradient $-\mathbf{U}$ and curvature $\mathbf{F}$. (i) Ridge ($\lambda_1 = 0$): the update solves the $(p+k)$ -dimensional system with $\lambda_2 \mathbf{I}$ added to the γ -block; all conclusions of Corollary 1 hold. (ii) Lasso/elastic net: one cycle of coordinate descent on the penalized quadratic costs $O((p+k)^2)$ —independent of N —after the pass, so the inner solver of Friedman et al. (2010) runs entirely on cell-collapsed quantities; the outer iteration is a proximal Newton method and converges quadratically to the penalized optimum near the solution under the conditions of Lee et al. (2014). (iii) The Karush–Kuhn–Tucker conditions, hence the active set and the regularization path, are functions of the tilted moments only.*

The practical consequence: a lasso path for a Poisson model with $N = 10^7$, $J = 10^3$ cells, and $p = 50$ expression covariates costs per path point what a dense implementation spends per coordinate.

5.3 Firth’s adjustment inside the tilted pass

Under separation (Section 3.4) the MLE does not exist; Firth’s bias-reducing penalty $\ell^*(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}) + \frac{1}{2} \log \det \mathbf{F}(\boldsymbol{\theta})$ (Firth, 1993) guarantees a finite maximizer for binomial models (Heinze and Schemper, 2002) and removes the $O(N^{-1})$ bias term in general. For canonical links the adjusted score has the closed form $\mathbf{U}^* = \mathbf{U} + \sum_i h_i \frac{b'''(\eta_i)}{2b''(\eta_i)} \mathbf{a}_i$ with leverages $h_i = \omega_i \mathbf{a}_i' \mathbf{F}^{-1} \mathbf{a}_i$ (Firth, 1993; Kosmidis and Firth, 2009).

Proposition 4 (Tilted Firth). *Given the partitioned inverse of $\mathbf{F}$ (available from the Newton solve), the leverages decompose as $h_i = \omega_i \{\mathbf{x}_i' \mathbf{V}_{\gamma\gamma} \mathbf{x}_i + 2 \mathbf{x}_i' \mathbf{c}_{j(i)} + d_{j(i)}\}$ with cell-level precomputations $\mathbf{c}_j = \mathbf{V}_{\gamma\delta} \mathbf{w}_j$ and $d_j = \mathbf{w}_j' \mathbf{V}_{\delta\delta} \mathbf{w}_j$ at $O(J(kp + k^2))$; the adjusted score, and hence the entire Firth quasi-scoring iteration (Kosmidis and Firth, 2009), is computable at $O(N(p^2 + p)) + O(J(k^2 + kp)) + O((p+k)^3)$ per iteration—the same order as Corollary 1. For Poisson, $b'''/2b'' \equiv 1/2$; for logit it is $\frac{1}{2}(1 - 2\mu_i)$.*

Thus the statistically principled response to separation costs essentially nothing beyond the unpenalized fit, removing the last reason to respond to separation by ad hoc data deletion. Section 6.4 quantifies the finite-sample behavior.

5.4 Robust and clustered variance estimation

All variance estimators used in practice share the cell structure. The observed/expected information is $\mathbf{F}$ itself (Theorem 1). The HC-type sandwich needs $\sum_i \hat{u}_i^2 \mathbf{a}_i \mathbf{a}_i'$ with $\hat{u}_i = y_i - \hat{\mu}_i$;

Table 1: Study 1: tilted-moment versus dense Newton. “Iterate dev.” is the maximum absolute difference between the two iterate sequences over all iterations and coordinates; times are medians per Newton iteration. Selected configurations; the full grid is in the supplement.

Family	N	J	p	Iterate dev.	Final dev.	t_{dense} (s)	t_{tilted} (s)
Poisson	2×10^5	50	2	5.1×10^{-14}	2.4×10^{-15}	0.119	0.010
Poisson	2×10^5	200	2	5.1×10^{-14}	3.7×10^{-15}	0.503	0.010
Poisson	2×10^5	500	2	6.0×10^{-13}	3.0×10^{-14}	1.481	0.017
Poisson	10^6	200	2	7.4×10^{-14}	1.1×10^{-14}	2.490	0.062
Poisson	2×10^5	200	5	1.1×10^{-13}	1.9×10^{-15}	0.520	0.016
Logistic	2×10^5	500	2	2.2×10^{-14}	4.0×10^{-15}	1.738	0.015
Logistic	10^6	200	2	8.2×10^{-15}	2.2×10^{-16}	2.918	0.082
Gamma	2×10^5	500	2	1.3×10^{-13}	5.2×10^{-15}	1.443	0.013
Gamma	10^6	200	2	2.5×10^{-13}	1.3×10^{-15}	2.662	0.058

its blocks accumulate cellwise exactly like (5) with ω_i replaced by $\hat{u}_i^2$. The cluster-robust estimator sums scores within clusters before the outer product; when clusters are unions of cells (the common design: cluster on region, cells within region) the accumulation is again one pass. These identities are what make *inference*, not just point estimation, run at cell-collapsed cost; they are implemented in the reference code and validated against dense computation in the test suite.

6 Simulation studies

All studies run on a single laptop-class core (Python/NumPy reference implementation; scripts `sim1–sim4` in the code supplement) and complete in minutes; every number below regenerates from a fixed seed.

6.1 Exactness and cost of tilted-moment Newton

Study 1 fits Poisson, logistic, and gamma models with $J \in \{50, 200, 500\}$ cells, $p \in \{2, 5\}$ continuous covariates, and N up to 10^6 , tracing the full iterate sequence of tilted-moment Newton and of textbook dense Newton from the same starting point. Table 1 reports the maximum deviation between the two iterate sequences across all iterations, the final-estimate deviation, and the per-iteration wall-clock times.

Three observations. First, the iterate sequences agree to 10^{-13} – 10^{-15} throughout—floating-point reordering noise—confirming Theorem 1(iv): this is the same algorithm, not an approximation. Second, measured per-iteration speedups range from $12\times$ ($J = 50$) to $117\times$ ($J = 500$), growing with k as Corollary 1 predicts, though below the raw FLOP ratios ($430\times$ – $4700\times$) because the dense baseline runs BLAS-3 while the tilted pass is memory-bound—the FLOP ratio is the right guide to scaling, the measured ratio to wall-clock budgeting. Third, the dense baseline at $N = 10^6$, $J = 200$ already needs a 1.6 GB design matrix where the tilted pass needs

Table 2: Study 2: predicted versus empirical per-sweep contraction of weighted alternating projections; sweeps to $\|\Delta\|_\infty < 10^{-13}$.

ρ	Predicted $c_F^2 = \sigma_2^2$	Empirical rate	Sweeps
0.00	0.0024	0.0019	7
0.30	0.1897	0.1873	19
0.60	0.5567	0.5547	48
0.80	0.7542	0.7534	96
0.90	0.8896	0.8747	197
0.95	0.9469	0.9469	468
0.99	0.9911	0.9910	2487

the p continuous columns only; at the dimensions of Section 7 the dense fit is not slow but impossible.

6.2 Convergence rates of alternating projections

Study 2 validates Theorem 3(ii) and Corollary 2. Two factors ($J_1 = 100$, $J_2 = 80$, $N = 2 \times 10^5$) are assigned with a mixing parameter ρ : with probability ρ a unit’s level of factor 2 is deterministically tied to its level of factor 1, otherwise uniform—so ρ sweeps the weighted bipartite graph from expander-like ($\rho = 0$) to nearly disconnected ($\rho = 0.99$). At Poisson optimum weights we compute the predicted per-sweep rate σ_2^2 from the normalized biadjacency kernel, then measure the empirical contraction factor of weighted alternating-projections demeaning of a generic covariate.

Table 2 shows agreement to the third decimal across the entire range, including the near-disconnected regime where the rate is within 10^{-3} of 1 and the sweep count grows as $(1 - c_F^2)^{-1}$. The spectral prediction is computable before iterating (a sparse SVD of the 100×80 weighted cross-tabulation) and could be used to warn users, select orderings, or switch to conjugate-gradient acceleration when the gap is small (Gaure, 2013; Correia, 2016).

6.3 The proportional regime

Study 3 probes Theorem 6 at its most extreme: Poisson cells of bounded size $m \in \{2, 5, 20\}$, one fixed effect per cell, heterogeneous cell effects $\alpha_j \sim \mathcal{N}(-1.2, 0.8^2)$, a covariate correlated with the cell effects (so that ignoring the effects biases $\hat{\gamma}$), $N \in \{20,000, 80,000\}$, 1,000 replications. With $m = 2$ the design estimates one nuisance parameter per two observations ($c = 1/2$).

Table 3 confirms every element of the theorem: bias is an order of magnitude below the Monte Carlo standard deviation even at $c = 1/2$ and shrinks with N at fixed c ; the profile-information standard errors track the true sampling variability to the third decimal; and both interval constructions cover at 94–96% throughout. For contrast, the same data analyzed with the cell effects ignored (a single intercept) biases the coefficient of the effect-correlated covariate

Table 3: Study 3: profile (concentrated) Poisson estimation with $J/N = 1/m$; 1,000 replications; $\gamma_{01} = 0.3$. “Cover” is the empirical coverage of nominal 95% intervals from the profile information (P) and the cell-clustered sandwich (C).

N	m	J/N	$\text{Bias}(\hat{\gamma}_1)$	$\text{SD}(\hat{\gamma}_1)$	$\overline{\text{SE}}_P$	Cover_P	Cover_C
20,000	2	0.50	−0.0003	0.0156	0.0157	0.954	0.953
20,000	5	0.20	+0.0009	0.0118	0.0119	0.953	0.951
20,000	20	0.05	+0.0004	0.0107	0.0107	0.954	0.953
80,000	2	0.50	+0.0004	0.0080	0.0078	0.953	0.951
80,000	5	0.20	−0.0001	0.0059	0.0060	0.955	0.954
80,000	20	0.05	−0.0001	0.0054	0.0053	0.943	0.941

Table 4: Study 4: quasi-separation frequency and estimator behavior; true $\gamma_z = 1.5$. MLE summaries condition on non-separated replications; Firth summaries use all replications. MAE is median absolute error.

N	Separated	Firth converged	$\text{med}(\hat{\gamma}_z^{\text{MLE}})$	$\text{med}(\hat{\gamma}_z^{\text{Firth}})$	MAE_{MLE}	$\text{MAE}_{\text{Firth}}$
300	0.394	1.000	1.704	1.586	0.587	0.498
600	0.042	1.000	1.581	1.556	0.446	0.400
1200	0.002	1.000	1.519	1.512	0.235	0.222
2400	0.000	1.000	1.520	1.515	0.199	0.195

by +0.28 (truth −0.2) and yields nominal 95% intervals with *zero* empirical coverage in every configuration—the point being that the fixed effects are not ignorable here, they are merely free.

6.4 Separation and the Firth-adjusted tilted pass

Study 4 stresses the boundary where Assumption 2 fails. Logistic model, $J = 10$ cells, a continuous covariate, and a rare binary exposure (3% prevalence, true log-odds 1.5): in small samples the exposed events can be quasi-separated with appreciable probability. We compare the MLE (tilted Newton, non-convergence or $|\hat{\gamma}_z| > 10$ flagged as separated) with the Firth-adjusted tilted iteration of Proposition 4, 500 replications per N .

At $N = 300$ the MLE fails to exist in 39% of replications; the Firth iteration converges in all 500, with smaller median bias *conditional on the MLE existing at all* and uniformly smaller median absolute error. As N grows the two estimators merge, as theory requires. The cost of the adjustment was a factor ~ 2 per iteration (the extra unadjusted pass for current-iterate leverages), consistent with Proposition 4.

7 Register-scale case study

We close with a case study at the scale that motivates the framework: spatial epidemiology on an emulated population register. Because genuine Nordic register microdata cannot leave

certified secure environments, the study uses a fully reproducible synthetic register, calibrated to published all-cause mortality gradients (Gompertz age slope, female advantage of ~ 0.45 on the log scale, educational gradients of 0.1–0.4, a secular decline of 1.2% per year); the generator is 40 lines of the supplement and every number below regenerates end to end in under a minute. The scientific template is a standard one—does a within-municipality environmental exposure predict mortality after fine demographic and spatial-temporal control?—and the computational shape is exactly that of Sections 2–3.

The data comprise $N = 5,000,000$ person-year episodes (35,526 deaths; crude rate 9.5 per 1,000 person-years) with: municipality $\times$ year fixed effects ($500 \times 20 = 10,000$ groups) absorbed by Algorithm 1; a cell-level design of education (4) $\times$ sex (2) $\times$ age band (12), $J = 96$ cells, $k = 16$ parameters; and $p = 2$ continuous covariates—standardized log income and a standardized pollution-exposure index correlated with the municipality effects (making the fixed effects non-ignorable for it). Standard errors are clustered on the 500 municipalities.

The fit: 322 zero-event municipality–year groups (159,464 episodes) are dropped by the separation rule; the remaining model—9,678 absorbed effects, 16 cell-level and 2 individual-level coefficients—converges in 8 tilted Newton iterations, 11.2 seconds end to end on one core. A dense dummy-variable design for the same model would occupy $N \times 9,696$ doubles ≈ 0.4 TB before the first iteration. For verification at feasible scale, the subpopulation of the first 10 municipalities ($n = 97,723$, 196 absorbed groups) was fitted both ways: maximum absolute coefficient deviation 4.0×10^{-11} , tilted 0.1s versus dense 3.0s.

Substantively, the model recovers the calibrated truth within sampling error: income -0.180 (cluster SE 0.005; truth -0.18), pollution $+0.049$ (SE 0.005; truth $+0.06$, within two SEs, identified from within-municipality variation only, as it must be after absorption), female -0.455 (truth -0.45), education gradients $-0.13/ -0.27/ -0.40$ (truth $-0.12/ -0.25/ -0.40$), and a monotone age gradient tracking the Gompertz truth to two decimals. The point of the exercise is not these numbers—the generator made them true—but the demonstration that a model of this shape, with inference, is an interactive-latency computation under the tilted-moment organization and an infeasible one under the dense organization; on genuine register data inside a secure environment, where wall-clock and memory budgets are tight and software dependencies restricted, this difference is what determines whether such models get fitted at all.

8 Discussion

This paper reframed two computational folk devices—cell collapse and fixed-effect absorption—as a single exact framework and supplied the theory that practice has been running ahead of: an algebraic identity (Theorem 1) that makes Newton on the full likelihood a cell-level computation; global convergence and sharp local rates for the nonlinear Gauss–Seidel absorption cycle (Theorems 2–3), with the rate identified as a bipartite spectral gap computable before

iterating (Corollary 2); and asymptotics that cover the dimensionalities register data actually present, including the proportional regime $J/N \rightarrow c$ where the Poisson profile likelihood’s coincidence with a conditional likelihood delivers unbiased, efficiently-normal, cluster-robust inference (Theorem 6).

Several limitations delineate future work. First, the proportional-regime theory is Poisson-specific; for the logit the conditional likelihood exists but is not the profile likelihood, and integrating conditional-ML or analytic bias corrections (Fernández-Val and Weidner, 2016) into the tilted pass is open. Second, Theorem 3 characterizes the local rate for $G = 2$ absorbed factors exactly; for $G \geq 3$ the block Gauss–Seidel radius bound is available but the clean angle characterization is not, matching the state of the linear alternating-projections literature (Deutsch, 2001). Third, our separation handling is exact for the absorbed blocks and delegated to Firth for the rest; a full polyhedral detection in the sense of Correia et al. (2019), run on cell aggregates, would be a useful complement. Fourth, the asymptotic regimes here treat cells and effects as fixed configurations; random-effects and crossed-random-effects limits under $J/N \rightarrow c$ raise different questions (Jochmans and Weidner, 2019).

Software. The algorithms are implemented twice. The Stata command `stpois` (with companion article) implements the Poisson event-history case in production form—`stset` integration, factor-variable parsing, weights, survey support—using the fast path (Theorem 1), the absorption path (Algorithm 1), and their combination. A self-contained reference implementation in Python/NumPy (roughly 600 lines, `tilted_glm.py`), which generated every number in Sections 6–7, accompanies this article; it implements all three families, absorption, Firth, and the robust/cluster variance estimators, and is organized to make the correspondence between code and theorems line-by-line auditable. Both are available at <https://github.com/andreasjungstrm/stpois>.

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A Proofs for Section 2

A.1 Proof of Proposition 1

(i) If $\mathbf{x}_i = \bar{\mathbf{x}}_j$ within cells, then $\eta_i = \boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j + \log t_i$ and (3) becomes $\sum_j \{(\boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j)D_j - T_j e^{\boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j}\} + \text{const}$, a function of the displayed cell statistics; the collapsed likelihood is the microdata likelihood up to an additive constant, so their maximizers and Hessians coincide.

(ii) For the Poisson family,

$$\ell(\gamma, \delta) = \sum_j \left\{ \gamma' S_j + \delta' \mathbf{w}_j D_j - e^{\delta' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\gamma) \right\} + \text{const},$$

which establishes computability from $\{S_j, D_j, \mathcal{L}_{\nu_j}\}$. Conversely, suppose a statistic permits evaluation of ℓ on an open set $\mathcal{O} \ni (\gamma_0, \delta_0)$. The functions $\delta \mapsto e^{\delta' \mathbf{w}_j}$ for distinct vectors $\mathbf{w}_j$ are linearly independent on any open set, so from $\delta \mapsto \sum_j e^{\delta' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\gamma)$ one recovers each $\mathcal{L}_{\nu_j}(\gamma)$ for each γ in the open slice. Each ν_j is a finite discrete measure, so $\mathcal{L}_{\nu_j}$ is entire; its values on an open subset of $\mathbb{R}^p$ determine it everywhere by analytic continuation, and the Laplace transform of a finite (positive) measure determines the measure. Hence the statistic determines ν_j , whose minimal description has dimension equal to (one plus p times) its support size.

(iii) Jensen's inequality applied to the strictly convex map $z \mapsto e^z$ under the probability measure ν_j/T_j gives $T_j^{-1} \mathcal{L}_{\nu_j}(\gamma) \geq \exp(\gamma' \bar{\mathbf{x}}_j)$ with equality iff $\gamma' \mathbf{x}$ is ν_j -a.s. constant. $\square$

A.2 Proof of Theorem 1

Differentiating (3) with $\theta_i = \eta_i$: $\phi \partial \ell / \partial \gamma = \sum_i (y_i - b'(\eta_i)) \mathbf{x}_i$ and $\phi \partial \ell / \partial \delta = \sum_i (y_i - b'(\eta_i)) \mathbf{w}_{j(i)}$. Grouping the second sum by cells and using that $\mathbf{w}_j$ is constant within cell j gives $\sum_j \mathbf{w}_j R_j$; grouping the first gives $\sum_i y_i \mathbf{x}_i - \sum_j \mathbf{m}_j^{b'}$ with $\mathbf{m}_j^{b'} = \sum_{i \in j} b'(\eta_i) \mathbf{x}_i = B_j \mathbb{E}_{P_{j, \theta}^{b'}}[\mathbf{x}]$ by the definition of the b' -tilt. Differentiating again, $\phi \mathbf{F} = \sum_i b''(\eta_i) \mathbf{a}_i \mathbf{a}_i'$ with $\mathbf{a}_i = (\mathbf{x}_i', \mathbf{w}_{j(i)}')'$; expanding the blocks and grouping by cells yields (ii), and the moment identities follow from Definition 1. For (iii), $b' = b'' = \exp$ gives $\omega_i = e^{\delta' \mathbf{w}_j} t_i e^{\gamma' \mathbf{x}_i}$, the cell factor cancels in the normalization, and $M_j^{(0)} = e^{\delta' \mathbf{w}_j} \mathcal{L}_{\nu_j}(\gamma)$; the cumulant identities $\mathbb{E}_{P_{j, \gamma}}[\mathbf{x}] = \nabla K_j$ and $\text{Var} = \nabla^2 K_j$ are the standard derivatives of a log-Laplace transform, and $\mathbb{E}[\mathbf{x} \mathbf{x}'] = \text{Var} + \mathbb{E} \mathbb{E}'$ gives the stated γ -block. (iv) The score and information assembled from (i)–(ii) are, entry for entry, the same matrices as those computed from the stacked design; hence so is the Newton step. $\square$

A.3 Proof of Corollary 1

The microdata pass evaluates, per observation, η_i from the cached cell linear predictor plus $\gamma' \mathbf{x}_i$ ($O(p)$), the scalars $b'(\eta_i), b''(\eta_i)$ ($O(1)$), the binned accumulations of ω_i and $\omega_i \mathbf{x}_i$ into cell $j(i)$ ($O(p)$), and the symmetric rank-one update $\omega_i \mathbf{x}_i \mathbf{x}_i'$ ($O(p^2)$ scalar multiply-adds, $p(p+1)/2$ distinct); total $O(N(p^2 + p + 1))$. Assembly of $\sum_j M_j^{(0)} \mathbf{w}_j \mathbf{w}_j'$ and $\sum_j \mathbf{m}_j \mathbf{w}_j'$ costs $O(J(k^2 + kp))$, and the symmetric solve $O((p+k)^3)$. The dense iteration forms $\sum_i \omega_i \mathbf{a}_i \mathbf{a}_i'$ at $O(N(p+k)^2)$ plus the same solve. Ratios follow. With $p = 0$ the pass reduces to two binned sums ($R_j, M_j^{(0)}$), and if the offsets enter through cell sums (pure rate models), even these are fixed after one pass. $\square$

A.4 Proof of Proposition 2

The ML score for a general link is $\sum_i (y_i - \mu_i) \frac{\partial \mu_i / \partial \eta_i}{V(\mu_i)} \mathbf{a}_i = \sum_i (y_i - \mu_i) g'(\mu_i) \omega_i \mathbf{a}_i = \sum_i r_i \mathbf{a}_i$, using $\partial \mu / \partial \eta = 1/g'(\mu)$, and the expected information is $\sum_i \omega_i \mathbf{a}_i \mathbf{a}_i'$ (McCullagh and Nelder, 1989, Sec. 2.5). These have the same cell-decomposable form as in Theorem 1 with (ω_i, r_i) in place of $(b''(\eta_i), y_i - b'(\eta_i))$, so the accumulation and complexity claims carry over. A fixed point of the scoring iteration has $\sum_i r_i \mathbf{a}_i = \mathbf{0}$, i.e., is a root of the ML score equations. For the gamma-log pair, $V(\mu) = \mu^2$ and $g'(\mu) = 1/\mu$ give $\omega_i = 1/(\mu_i^2 \mu_i^{-2}) = 1$; the moments (5) are then constant in $\boldsymbol{\theta}$, and only $R_j = \sum_{i \in j} (y_i - \mu_i) / \mu_i$ requires re-evaluation, an $O(Np)$ pass. $\square$

B Proofs for Section 3

Throughout this appendix write $D = q + \sum_g J_g$ for the total parameter dimension, $\mathbf{a}_i \in \mathbb{R}^D$ for the stacked design vector of observation i (individual-level, cell-level, and absorbed indicator parts), $\mathcal{K} = \{\mathbf{u} : \mathbf{a}_i' \mathbf{u} = 0 \ \forall i\}$ for the null space, and $E = \mathcal{K}^\perp \cong \mathbb{R}^D / \mathcal{K}$. By Assumption 1, $\ell(\boldsymbol{\theta}) = \phi^{-1} \sum_i \{y_i \mathbf{a}_i' \boldsymbol{\theta} - b(\mathbf{a}_i' \boldsymbol{\theta} + o_i)\}$ (absorbing offsets into b 's argument) is concave on $\mathbb{R}^D$, constant on cosets of $\mathcal{K}$, and its restriction $\bar{\ell}$ to E is *strictly* concave: for $\mathbf{u} \in E \setminus \{\mathbf{0}\}$, some $\mathbf{a}_i' \mathbf{u} \neq 0$ and $-\mathbf{u}' \nabla^2 \ell \mathbf{u} = \phi^{-1} \sum_i b''(\cdot)(\mathbf{a}_i' \mathbf{u})^2 > 0$.

Lemma 2 (Compact superlevel sets). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be strictly concave, continuous, and attain its maximum at x^* . Then every superlevel set $\{x : f(x) \geq a\}$, $a \leq f(x^*)$, is compact.*

Proof. Superlevel sets of concave functions are closed and convex. Suppose S_a is unbounded for some a ; then S_a contains a ray $\{x^* + t\mathbf{v} : t \geq 0\}$ for some unit vector $\mathbf{v}$ (a closed unbounded convex set containing x^* contains a ray from x^* ; see Rockafellar, 1970, Thm. 8.4). Let $\varphi(t) = f(x^* + t\mathbf{v})$: strictly concave on $[0, \infty)$, $\varphi(t) \geq a$ for all t , maximized at $t = 0$. If $\varphi'(t_0^+) < 0$ for some t_0 , concavity gives $\varphi(t) \leq \varphi(t_0) + \varphi'(t_0^+)(t - t_0) \rightarrow -\infty$, contradicting $\varphi \geq a$; hence the (strictly decreasing, by strict concavity) right derivative satisfies $\varphi'(t^+) \geq 0$ for all $t > 0$. But $\varphi'(0^+) \leq 0$ at the maximum, and strict decrease then forces $\varphi'(t^+) < 0$ for $t > 0$ —a contradiction. $\square$

B.1 Proof of Theorem 2

All statements are on E ; representatives are normalized after each sweep (any fixed measurable normalization, e.g., recentering each absorbed factor and compensating in the intercept, changes nothing in ℓ).

Monotonicity and confinement. Each $\boldsymbol{\alpha}^{(g)}$ -update is an exact maximization over its block (the block objective separates over the levels of factor g ; each level's scalar problem $\alpha \mapsto c_1 \alpha - \sum_i b(\alpha + \text{rest}_i)$ is strictly concave with derivative $\rightarrow \pm\infty$ prevented only if the level has an interior event, which Assumption 3(i) guarantees; hence a unique finite maximizer exists).

The β -update is accepted only under the Armijo test, so $\ell(\boldsymbol{\theta}_{t+1}) \geq \ell(\boldsymbol{\theta}_t)$, and the sequence $\ell(\boldsymbol{\theta}_t)$ converges to some $L^* \leq \ell(\boldsymbol{\theta}^*)$ (Assumption 2). By Lemma 2 applied to $\bar{\ell}$ on E , the normalized iterates remain in the compact set $\bar{S} = \{\bar{\ell} \geq \ell(\boldsymbol{\theta}_0)\}$.

Limit points are stationary. Let $\bar{\boldsymbol{\theta}}$ be a limit point of the normalized iterates, along t_r . The per-cycle increment $\ell(\boldsymbol{\theta}_{t+1}) - \ell(\boldsymbol{\theta}_t) \rightarrow 0$, and increments are sums of nonnegative per-block increments, so each block's increment vanishes. Fix a block g and let T_g denote the map that exactly maximizes over block g holding the rest fixed; T_g is continuous on $\bar{S}$ (the unique maximizer of a strictly concave, jointly continuous objective depends continuously on the fixed coordinates; Berge's maximum theorem plus uniqueness). If block g were not optimal at $\bar{\boldsymbol{\theta}}$, then $\varepsilon := \ell(T_g \bar{\boldsymbol{\theta}}) - \ell(\bar{\boldsymbol{\theta}}) > 0$; by continuity of T_g and ℓ , the block- g increment at $\boldsymbol{\theta}_{t_r}$ (with the preceding blocks of the cycle already updated, which also converge along the subsequence by the same argument applied inductively in the cycle order) exceeds $\varepsilon/2$ for large r —contradiction. Hence every absorbed block of $\bar{\boldsymbol{\theta}}$ is blockwise optimal, i.e., $\nabla_{\alpha} \ell(\bar{\boldsymbol{\theta}}) = \mathbf{0}$. For the β -block, suppose $\mathbf{U}_{\beta}(\bar{\boldsymbol{\theta}}) \neq \mathbf{0}$. On the compact set $\bar{S}$, the information $\mathbf{F}$ is continuous and, by Assumption 3(ii), $\mathbf{F}_{\beta\beta}$ is nonsingular with eigenvalues in a fixed interval $[\underline{\lambda}, \bar{\lambda}]$, $\underline{\lambda} > 0$; the Newton direction $\mathbf{d} = \mathbf{F}_{\beta\beta}^{-1} \mathbf{U}_{\beta}$ then satisfies $\mathbf{U}_{\beta}' \mathbf{d} \geq \|\mathbf{U}_{\beta}\|^2 / \bar{\lambda}$, and the standard backtracking bound guarantees an accepted step length λ_t bounded below and an increment $\geq \sigma \lambda_t \mathbf{U}_{\beta}' \mathbf{d}$ bounded away from zero in a neighborhood of $\bar{\boldsymbol{\theta}}$ (Bertsekas, 1999, Prop. 1.2.1 and Sec. 2.7)—again contradicting vanishing increments. Hence $\nabla \ell(\bar{\boldsymbol{\theta}}) = \mathbf{0}$.

Conclusion. For a concave differentiable function, a stationary point is a global maximizer, so $\bar{\boldsymbol{\theta}} = \boldsymbol{\theta}^*$ modulo $\mathcal{K}$ by strict concavity on E ; since every limit point of the normalized iterates in the compact set $\bar{S}$ equals the normalized $\boldsymbol{\theta}^*$, the whole sequence converges to it, and $L^* = \ell(\boldsymbol{\theta}^*)$, proving (i)–(ii). Part (iii) follows because the argument used only: exactness/uniqueness of the absorbed block maximizations, sufficient ascent of the β -step, and vanishing per-block increments—all invariant to replacing the β -step by exact maximization and to essentially-cyclic orderings (every block updated at least once every B iterations). $\square$

B.2 Proof of Theorem 3

(i) Write the inner objective as $f(\boldsymbol{\alpha}) = \ell(\boldsymbol{\beta}, \boldsymbol{\alpha})$ with fixed $\boldsymbol{\beta}$, and let $\hat{\boldsymbol{\alpha}}$ be its maximizer (unique after normalization). The block- g exact update solves $\nabla_g f(\boldsymbol{\alpha}^{(<g),+}, \boldsymbol{\alpha}^{(g)}, \boldsymbol{\alpha}^{(>g)}) = \mathbf{0}$; the block Hessian $\mathbf{F}_{gg}$ is diagonal with entries $\sum_{i \in \text{level}} b''(\eta_i) > 0$, so by the implicit function theorem (using $b \in C^3$) the update map is C^1 near the fixed point with partial derivatives $-\mathbf{F}_{gg}^{-1} \mathbf{F}_{g,h}$ with respect to block $h \neq g$. Composing the G block updates in cycle order gives $DT(\hat{\boldsymbol{\alpha}}) = -(\mathbf{D} + \mathbf{L})^{-1} \mathbf{L}'$, where $\mathbf{F}_{\alpha\alpha} = \mathbf{D} + \mathbf{L} + \mathbf{L}'$ is the block diagonal/lower/upper decomposition—the classical Gauss–Seidel error operator—restricted to the quotient. Since $\mathbf{F}_{\alpha\alpha}$ is symmetric positive definite on the quotient, $\rho_{\text{GS}} = \rho(DT) < 1$ (Golub and Van Loan, 2013, Sec. 11.2), and Ostrowski's theorem (Ortega and Rheinboldt, 1970, Sec. 10.1) yields local R -linear convergence at rate ρ_{GS} .

(ii) Let $G = 2$ with indicator matrices $\mathbf{A}_1 \in \mathbb{R}^{N \times J_1}$, $\mathbf{A}_2 \in \mathbb{R}^{N \times J_2}$ and $\mathbf{F}_{gh} = \mathbf{A}_g' \Omega^* \mathbf{A}_h$. The eigenvalues of the two-block Gauss–Seidel operator are $\{0\}$ together with the eigenvalues of $\mathbf{F}_{22}^{-1} \mathbf{F}_{21} \mathbf{F}_{11}^{-1} \mathbf{F}_{12}$ (eliminate the first block). In the Ω^* inner product this matrix represents $\mathbf{P}_2 \mathbf{P}_1|_{V_2}$, the compression to V_2 of the product of the orthogonal projections onto V_1 and V_2 ; its spectrum on the complement of $V_1 \cap V_2$ is $\{\cos^2 \theta_i\}$ with θ_i the principal angles between $V_1 \ominus (V_1 \cap V_2)$ and $V_2 \ominus (V_1 \cap V_2)$, whose supremum-cosine is c_F (Deutsch, 2001, Ch. 9). On the quotient (which removes $V_1 \cap V_2$, the flat directions), $\rho = c_F^2$. The identical rate for the demeaning residuals is the theorem of Aronszajn (1950) and Kayalar and Weinert (1988): $\|(\mathbf{P}_{V_2^\perp} \mathbf{P}_{V_1^\perp})^n - \mathbf{P}_{(V_1+V_2)^\perp}\|_{\Omega^*} = c_F^{2n-1}$, using that the Friedrichs angle is invariant under orthogonal complementation (Deutsch, 2001, Ch. 9).

(iii) Append the β -block: the cycle map with exact maximizations is again C^1 by the implicit function theorem (the β -block Hessian is nonsingular by Assumption 3(ii)), its Jacobian is the block Gauss–Seidel operator of the full $\mathbf{F}(\theta^*)$ in the stated ordering, and the SPD argument of (i) applies on the quotient. $\square$

B.3 Proof of Corollary 2

Represent $u \in V_1$, $v \in V_2$ by their level vectors, $u = \mathbf{A}_1 \tilde{u}$, $v = \mathbf{A}_2 \tilde{v}$. Then $\langle u, v \rangle_{\Omega^*} = \tilde{u}' \mathbf{B} \tilde{v}$, $\|u\|_{\Omega^*}^2 = \tilde{u}' \mathbf{D}_1 \tilde{u}$, $\|v\|_{\Omega^*}^2 = \tilde{v}' \mathbf{D}_2 \tilde{v}$. Substituting $\mathbf{r} = \mathbf{D}_1^{1/2} \tilde{u}$, $\mathbf{s} = \mathbf{D}_2^{1/2} \tilde{v}$ turns the Friedrichs supremum into the second singular value of $\mathbf{T} = \mathbf{D}_1^{-1/2} \mathbf{B} \mathbf{D}_2^{-1/2}$: the top singular pair $(\mathbf{D}_1^{1/2} \mathbf{1}, \mathbf{D}_2^{1/2} \mathbf{1})$, with $\sigma_1 = 1$, spans exactly the image of the intersection $V_1 \cap V_2 \supseteq \text{constants}$, which the Friedrichs angle excludes. $\mathbf{T}$ is the transition kernel of the two-step random walk on the weighted bipartite graph with biadjacency $\mathbf{B}$; by Perron–Frobenius, the multiplicity of $\sigma = 1$ equals the number of connected components, so $\sigma_2 < 1$ iff the graph is connected, which is also exactly the condition $V_1 \cap V_2 = \text{span}\{\mathbf{1}\}$ (Abowd et al., 1999; Gaure, 2013). $\square$

C Proofs for Section 4

C.1 Proof of Theorem 4

Under Assumption 4, the normalizing regularity and divergence conditions of Fahrmeir and Kaufmann (1985, Sec. 3) hold for the canonical-link exponential family with bounded covariates: their normed information $\mathbf{F}_N^{-1/2}$ exists for large N , and their Theorems 1–2 give existence, consistency, and $\mathbf{F}_N^{1/2}(\hat{\theta}_N - \theta_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \phi \mathbf{I})$; Assumption 4(ii) converts this to the stated $\sqrt{N}$ form. Efficiency is the standard exponential-family Cramér–Rao attainment. Under correct mean but misspecified distribution, the estimator is the PML estimator of Gouriéroux et al. (1984) for the linear exponential family, whose Theorem 3 gives the sandwich limit; consistency of the cell-accumulated sandwich estimator follows from the law of large numbers applied to $\hat{u}_i^2$ -weighted design moments, using the uniform consistency of $\hat{\mu}_i$ on bounded designs. $\square$

C.2 Proof of Theorem 5

Normalize $\phi = 1$. Write $\mathbf{F} = \mathbf{F}_N(\boldsymbol{\theta}_0)$ (positive definite for large N by Assumption 5(ii), which presupposes invertibility), $\mathbf{U} = \mathbf{U}(\boldsymbol{\theta}_0)$, and change variables $\mathbf{s} = \mathbf{F}^{1/2}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)$, $g(\mathbf{s}) = \ell(\boldsymbol{\theta}_0 + \mathbf{F}^{-1/2}\mathbf{s})$. Then g is concave, $\nabla g(\mathbf{0}) = \mathbf{F}^{-1/2}\mathbf{U}$, and $\nabla^2 g(\mathbf{s}) = -\mathbf{F}^{-1/2}\mathbf{H}(\boldsymbol{\theta}_s)\mathbf{F}^{-1/2}$.

Step 1: score bound. $\mathbb{E}\|\nabla g(\mathbf{0})\|^2 = \text{tr}(\mathbf{F}^{-1/2} \text{Var}(\mathbf{U}) \mathbf{F}^{-1/2}) = q_N$ by the information equality, so $\|\nabla g(\mathbf{0})\| = O_p(\sqrt{q_N})$.

Step 2: uniform curvature control. Fix $r \geq 1$ and let $\|\mathbf{s}\| \leq r\sqrt{q_N}$. By Assumption 5(i,iv), $\omega_i = b''(\eta_{0i}) \in [\underline{\omega}, \bar{\omega}]$ with $0 < \underline{\omega} \leq \bar{\omega} < \infty$, and the balanced-leverage condition gives $\|\mathbf{F}^{-1/2}\mathbf{a}_i\|^2 = h_i/\omega_i \leq Cq_N/(N\underline{\omega})$. Hence

$$\max_i |\eta_i(\boldsymbol{\theta}_s) - \eta_{0i}| = \max_i |\mathbf{a}_i' \mathbf{F}^{-1/2} \mathbf{s}| \leq r\sqrt{q_N} \cdot \sqrt{Cq_N/(N\underline{\omega})} = C_r q_N/\sqrt{N} =: \varepsilon_N \rightarrow 0$$

using $q_N^2/N \rightarrow 0$. With L the Lipschitz constant of b'' on the relevant compact set, $|\sum_i \{b''(\eta_i(\boldsymbol{\theta}_s)) - b''(\eta_{0i})\}(\mathbf{a}_i' \mathbf{u})^2| \leq (L\varepsilon_N/\underline{\omega}) \mathbf{u}' \mathbf{F} \mathbf{u}$ for all $\mathbf{u}$, so uniformly on the ball,

$$-(1 + \delta_N) \mathbf{I} \preceq \nabla^2 g(\mathbf{s}) \preceq -(1 - \delta_N) \mathbf{I}, \quad \delta_N := L\varepsilon_N/\underline{\omega} \rightarrow 0. \quad (8)$$

Step 3: existence and rate. On $\|\mathbf{s}\| = r\sqrt{q_N}$, a second-order Taylor expansion with (8) gives $g(\mathbf{s}) - g(\mathbf{0}) \leq \|\nabla g(\mathbf{0})\| r\sqrt{q_N} - \frac{1}{2}(1 - \delta_N)r^2 q_N$. On the event $\{\|\nabla g(\mathbf{0})\| \leq K\sqrt{q_N}\}$, whose probability exceeds $1 - K^{-2}\mathbb{E}\|\nabla g(\mathbf{0})\|^2/q_N = 1 - K^{-2}$, the right side is negative for $r > 4K$; concavity of g then confines the (unique, by strict concavity) maximizer $\hat{\mathbf{s}}$ to the open ball, proving existence and $\|\hat{\mathbf{s}}\| = \|\mathbf{F}^{1/2}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\| = O_p(\sqrt{q_N})$.

Step 4: asymptotic linearity. By the integral form of the mean value theorem, $\mathbf{0} = \nabla g(\hat{\mathbf{s}}) = \nabla g(\mathbf{0}) - \tilde{\mathbf{J}}\hat{\mathbf{s}}$ with $\tilde{\mathbf{J}} = -\int_0^1 \nabla^2 g(t\hat{\mathbf{s}}) dt$; by (8), $\|\tilde{\mathbf{J}} - \mathbf{I}\| \leq \delta_N$, so

$$\|\hat{\mathbf{s}} - \nabla g(\mathbf{0})\| \leq \frac{\delta_N}{1 - \delta_N} \|\nabla g(\mathbf{0})\| = O_p\left(\frac{q_N}{\sqrt{N}}\sqrt{q_N}\right) = O_p(q_N^{3/2}/\sqrt{N}) = o_p(1)$$

under $q_N^3/N \rightarrow 0$.

Step 5: contrasts. For $\mathbf{C}_N$ with $\boldsymbol{\Sigma}_{C,N} = \mathbf{C}_N' \mathbf{F}^{-1} \mathbf{C}_N \rightarrow \boldsymbol{\Sigma}_C \succ \mathbf{0}$ and any fixed $\mathbf{v} \in \mathbb{R}^d$, write $Z_N = \mathbf{v}' \boldsymbol{\Sigma}_{C,N}^{-1/2} \mathbf{C}_N' (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \mathbf{v}' \boldsymbol{\Sigma}_{C,N}^{-1/2} \mathbf{C}_N' \mathbf{F}^{-1/2} \hat{\mathbf{s}}$. The remainder from Step 4 contributes $\leq \|\boldsymbol{\Sigma}_{C,N}^{-1/2} \mathbf{C}_N' \mathbf{F}^{-1/2}\| o_p(1) = o_p(1)$ (the matrix has orthonormal-bounded rows by construction). The main term is $\sum_i c_{N,i}(y_i - b'(\eta_{0i}))$ with $c_{N,i} = \mathbf{v}' \boldsymbol{\Sigma}_{C,N}^{-1/2} \mathbf{C}_N' \mathbf{F}^{-1} \mathbf{a}_i$: independent, mean-zero, total variance $\mathbf{v}' \mathbf{v}(1 + o(1))$, and $\max_i |c_{N,i}| \leq \|\mathbf{v}\| \|\boldsymbol{\Sigma}_{C,N}^{-1/2}\| \|\mathbf{C}_N' \mathbf{F}^{-1/2}\| \max_i \|\mathbf{F}^{-1/2} \mathbf{a}_i\| = O(\sqrt{q_N/N}) \rightarrow 0$; since the residuals have uniformly bounded $(2 + \epsilon)$ moments on compact η -sets, the Lindeberg condition holds and $Z_N \xrightarrow{d} \mathcal{N}(0, \|\mathbf{v}\|^2)$; Cramér–Wold completes the claim. Consistency of the plug-in variance follows from (8) evaluated at $\hat{\mathbf{s}}$: $\|\mathbf{F}^{-1/2} \mathbf{F}(\hat{\boldsymbol{\theta}}) \mathbf{F}^{-1/2} - \mathbf{I}\| \leq \delta_N' \xrightarrow{p} 0$, which implies relative consistency of $\mathbf{C}_N' \mathbf{F}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{C}_N$. $\square$

C.3 Proof of Lemma 1

For fixed γ the inner maximization separates over cells; $\partial \ell / \partial \alpha_j = D_j - e^{\alpha_j} E_j(\gamma)$ with $E_j(\gamma) = \sum_i e^{\gamma' \mathbf{x}_{ij} + o_{ij}}$, giving $e^{\hat{\alpha}_j} = D_j / E_j$ for $D_j > 0$ (for $D_j = 0$ the supremum is approached as $\alpha_j \rightarrow -\infty$ and the cell contributes a γ -free constant). Substituting,

$$\ell_p(\gamma) = \sum_{j: D_j > 0} \left\{ \sum_i y_{ij}(\gamma' \mathbf{x}_{ij} + o_{ij}) - D_j \log E_j(\gamma) \right\} + \sum_j D_j (\log D_j - 1).$$

The bracket equals $\sum_i y_{ij} \log \pi_{ij}(\gamma)$ plus terms free of γ , which is the multinomial log likelihood given D_j up to the combinatorial constant; and the multinomial law given the total is the standard Poisson-splitting identity. Concavity: $\gamma \mapsto \log E_j(\gamma)$ is log-sum-exp of affine maps, hence convex. Differentiating the bracket gives the stated score and, again, negative Hessian $\sum_j D_j \text{Var}_{\pi_j}(\mathbf{x}_j)$; the Schur form follows by evaluating (5) at $\hat{\alpha}(\gamma)$, where $\mu_{ij} = D_j \pi_{ij}$, so $M_j^{(0)} = D_j$, $\mathbf{m}_j = D_j \mathbb{E}_{\pi_j}[\mathbf{x}_j]$, and $\mathbf{Q} - \sum_j \mathbf{m}_j \mathbf{m}_j' / M_j^{(0)} = \sum_j D_j \{ \mathbb{E}_{\pi_j}[\mathbf{x} \mathbf{x}'] - \mathbb{E}_{\pi_j}[\mathbf{x}] \mathbb{E}_{\pi_j}[\mathbf{x}]' \}$. $\square$

C.4 Proof of Theorem 6

Throughout, expectations condition on the (fixed) covariates, offsets, and cell effects; cells are independent. Let $\ell_j(\gamma)$ denote cell j 's conditional (multinomial) log likelihood and $L_J(\gamma) = J^{-1} \sum_j \{ \ell_j(\gamma) - \ell_j(\gamma_0) \}$, so that $\hat{\gamma}_J$ maximizes the concave function L_J by Lemma 1.

(i) *Consistency.* Each summand satisfies, on the compact set Γ , $|\ell_j(\gamma) - \ell_j(\gamma_0)| \leq 2D_j \sup_{i, \gamma \in \Gamma} |(\gamma - \gamma_0)' \mathbf{x}_{ij}| \leq C_\Gamma D_j$, and by Assumption 6(ii) the cell means are bounded, so D_j has uniformly bounded moments of all orders; hence $\text{Var}\{L_J(\gamma)\} \leq C/J \rightarrow 0$ and $L_J(\gamma) - \mathbb{E}L_J(\gamma) \xrightarrow{p} 0$ pointwise. The mean is a per-cell Kullback–Leibler divergence: $\mathbb{E}\{\ell_j(\gamma) - \ell_j(\gamma_0)\} = -\mathbb{E} \text{KL}(\pi_j(\gamma_0) \parallel \pi_j(\gamma)) \leq 0$, and a second-order expansion at γ_0 with Assumption 6(iii) gives an $\epsilon_0 > 0$ and, for large J , $\mathbb{E}L_J(\gamma) \leq -\frac{\lambda}{4} \|\gamma - \gamma_0\|^2$ whenever $\|\gamma - \gamma_0\| \leq \epsilon_0$; concavity of $\mathbb{E}L_J$ (with $\mathbb{E}L_J(\gamma_0) = 0$) extends the strict separation to all of Γ : for $\|\gamma - \gamma_0\| \geq \epsilon$ with $\epsilon \leq \epsilon_0$, evaluating along the segment from γ_0 yields $\mathbb{E}L_J(\gamma) \leq -\frac{\lambda \epsilon^2}{4}$. By the convexity lemma (Andersen and Gill, 1982; Pollard, 1991; Niemiro, 1992), pointwise convergence of the concave functions $L_J - \mathbb{E}L_J$ to 0 is automatically uniform on compacts, and the argmax of L_J converges in probability to γ_0 .

(ii) *Normality.* The profile score at the truth is $\mathbf{U}_J(\gamma_0) = \sum_j \mathbf{s}_j$ with $\mathbf{s}_j = \sum_i (y_{ij} - D_j \pi_{ij}(\gamma_0)) \mathbf{x}_{ij}$. Conditionally on D_j the multinomial model is correctly specified, so $\mathbb{E}[\mathbf{s}_j \mid D_j] = \mathbf{0}$, hence $\mathbb{E} \mathbf{s}_j = \mathbf{0}$ exactly, for every J and every configuration of cell effects—this is the absence of incidental-parameter bias. The summands are independent with $\text{Var}(\mathbf{s}_j) = \mathbb{E}[D_j \text{Var}_{\pi_j}(\mathbf{x}_j)]$ (conditional information equality), so $\text{Var}(J^{-1/2} \mathbf{U}_J) = \bar{\mathbf{F}}_J(\gamma_0)$, and $\|\mathbf{s}_j\| \leq 2D_j \max_i \|\mathbf{x}_{ij}\|$ has bounded third moments, so Lyapunov's CLT gives $J^{-1/2} \mathbf{U}_J \xrightarrow{d} \mathcal{N}(\mathbf{0}, \bar{\mathbf{F}})$ by Assumption 6(iii). For the Hessian, $J^{-1} \mathbf{H}_p(\gamma) = J^{-1} \sum_j D_j \text{Var}_{\pi_j(\gamma)}(\mathbf{x}_j)$ is an average of independent terms, bounded by CD_j and Lipschitz in γ with constant $\leq C'D_j$ on Γ ; a standard bracketing/finite-

cover argument gives $\sup_{\|\gamma - \gamma_0\| \leq \epsilon_J} \|J^{-1} \mathbf{H}_p(\gamma) - \bar{\mathbf{F}}_J(\gamma_0)\| \xrightarrow{p} 0$ for any $\epsilon_J \rightarrow 0$. A Taylor expansion of the score around γ_0 , solved at $\hat{\gamma}_J$, then yields $\sqrt{J}(\hat{\gamma}_J - \gamma_0) = \bar{\mathbf{F}}_J^{-1} J^{-1/2} \mathbf{U}_J + o_p(1)$ and the stated limit.

(iii) *Variance estimation.* Consistency of $J^{-1} \mathbf{H}_p(\hat{\gamma}_J)$ for $\lim \bar{\mathbf{F}}_J$ is the uniform convergence just shown plus (i). For the clustered estimator, $J^{-1} \sum_j \mathbf{s}_j(\hat{\gamma}) \mathbf{s}_j(\hat{\gamma})'$: at γ_0 the LLN with bounded fourth moments gives convergence to $\lim J^{-1} \sum_j \text{Var}(\mathbf{s}_j)$; the difference between evaluation at $\hat{\gamma}$ and at γ_0 is bounded by a Lipschitz-in- γ envelope with integrable constant times $\|\hat{\gamma} - \gamma_0\| \xrightarrow{p} 0$. If the within-cell joint distribution is misspecified but the conditional mean $\mathbb{E}[y_{ij} \mid D_j] = D_j \pi_{ij}(\gamma_0)$ is correct, then $\mathbb{E} \mathbf{s}_j = \mathbf{0}$ still holds, the same argument applies with $\text{Var}(\mathbf{s}_j)$ no longer equal to the conditional information, and the sandwich $\mathbf{H}_p^{-1}(\sum_j \mathbf{s}_j \mathbf{s}_j') \mathbf{H}_p^{-1}$ remains consistent; the $G/(G-1)$ correction is asymptotically negligible.

(iv) *Efficiency.* In the model with unrestricted $(\alpha_1, \alpha_2, \dots)$, the nuisance tangent space at (γ_0, α_0) is spanned, cell by cell, by the α_j -scores $u_j = \sum_i (y_{ij} - \mu_{ij})$. The efficient score for γ is the residual of the projection of the γ -score onto this space; by independence across cells the projection is cellwise, and within cell j , $\mathbb{E}[(\sum_i (y_{ij} - \mu_{ij}) \mathbf{x}_{ij}) u_j] = \sum_i \mu_{ij} \mathbf{x}_{ij}$, $\mathbb{E}[u_j^2] = \sum_i \mu_{ij}$, so the projection residual is $\sum_i (y_{ij} - \mu_{ij}) \mathbf{x}_{ij} - \frac{\sum_i \mu_{ij} \mathbf{x}_{ij}}{\sum_i \mu_{ij}} \sum_i (y_{ij} - \mu_{ij}) = \mathbf{s}_j$ evaluated at the truth: exactly the conditional score. The asymptotic variance in (ii) equals the inverse of the efficient information, so the bound is attained (van der Vaart, 1998, Ch. 25). $\square$

D Proofs for Section 5

D.1 Proof of Proposition 3

(i) Adding $\frac{\lambda_2}{2} \|\gamma\|^2$ contributes $\lambda_2 \mathbf{I}_p$ to the γ -block of the curvature and $-\lambda_2 \gamma$ to its gradient; the modified system has the same dimensions and cost. (ii) The proximal Newton step minimizes the penalized quadratic model $m(\theta) = -\mathbf{U}'(\theta - \theta_t) + \frac{1}{2}(\theta - \theta_t)' \mathbf{F}(\theta - \theta_t) + \lambda_1 \|\gamma\|_1$ (plus the ridge term); one coordinate update needs the corresponding row of $\mathbf{F}$ and the current residual of the linear system, both dense in $\mathbb{R}^{p+k}$, so a full cycle costs $O((p+k)^2)$ —all references to data enter through $(\mathbf{U}, \mathbf{F})$, which Theorem 1 provides from one pass. Since $-\ell$ is convex with locally Lipschitz Hessian and $\mathbf{F} \succ \mathbf{0}$ near the solution, the inexact proximal Newton method converges locally quadratically (Lee et al., 2014, Thms. 3.4–3.5). (iii) The KKT conditions are $|\mathbf{U}]_r - [\mathbf{F}(\theta - \theta_t)]_r| \leq \lambda_1$ for inactive γ_r , with equality sign structure for active ones: functions of $(\mathbf{U}, \mathbf{F})$ only. $\square$

D.2 Proof of Proposition 4

For canonical links, $\mathbf{F}(\theta) = \sum_i b''(\eta_i) \mathbf{a}_i \mathbf{a}_i'$ depends on θ through the η_i , and $\partial \mathbf{F} / \partial \theta_r = \sum_i b'''(\eta_i) a_{ir} \mathbf{a}_i \mathbf{a}_i'$. Firth's adjustment (Firth, 1993) is $A_r = \frac{1}{2} \text{tr}(\mathbf{F}^{-1} \partial \mathbf{F} / \partial \theta_r) = \frac{1}{2} \sum_i b'''(\eta_i) a_{ir} \mathbf{a}_i' \mathbf{F}^{-1} \mathbf{a}_i = \sum_i h_i \frac{b'''(\eta_i)}{2b''(\eta_i)} a_{ir}$ with $h_i = b''(\eta_i) \mathbf{a}_i' \mathbf{F}^{-1} \mathbf{a}_i$: the adjusted score is the ordinary score with work-

ing residual $y_i - b'(\eta_i) + h_i b'''(\eta_i)/2b''(\eta_i)$, which the cell accumulation of Theorem 1 ingests unchanged. For the leverages, write the partitioned inverse blocks $\mathbf{V}_{\gamma\gamma}, \mathbf{V}_{\gamma\delta}, \mathbf{V}_{\delta\delta}$ of $\mathbf{F}^{-1}$ and expand $\mathbf{a}_i = (\mathbf{x}'_i, \mathbf{w}'_{j(i)})'$: $\mathbf{a}'_i \mathbf{F}^{-1} \mathbf{a}_i = \mathbf{x}'_i \mathbf{V}_{\gamma\gamma} \mathbf{x}_i + 2\mathbf{x}'_i (\mathbf{V}_{\gamma\delta} \mathbf{w}_{j(i)}) + \mathbf{w}'_{j(i)} \mathbf{V}_{\delta\delta} \mathbf{w}_{j(i)}$. Precomputing $\mathbf{c}_j = \mathbf{V}_{\gamma\delta} \mathbf{w}_j$ ($O(Jkp)$) and $d_j = \mathbf{w}'_j \mathbf{V}_{\delta\delta} \mathbf{w}_j$ ($O(Jk^2)$) leaves per-datum cost $O(p^2 + p)$, giving the stated totals; $b'''/2b''$ equals $1/2$ for Poisson and $(1 - 2\mu_i)/2$ for the logit by direct computation. The quasi-scoring iteration and its properties are those of Kosmidis and Firth (2009); finiteness of the penalized maximizer for binomial responses is Theorem 1 of Heinze and Schemper (2002). $\square$

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